FEL-1 Design and Techno-Economic Analysis for a Proposed Addition of a DMC Production Process to a Chemical Plant

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Authored by Design Team 9:

Lauren Bajo-Smith, Process Development Engineer

Forrest Ramirez, Process Development Engineer

Ashley Uriostegui, Process Development Engineer

Executive Summary

This report details our team's FEL-1 design of a proposed oxidative carbonylation process to produce dimethyl carbonate (sold for $900 per MT) as an addition to a chemical plant. For this process, methanol (purchased at $400 per MT), carbon monoxide (purchased at $150 per MT), and oxygen gas (purchased at $100 per MT) undergo an oxidative carbonylation reaction to produce primary product dimethyl carbonate and co-product water. This reaction has the potential to become a runaway reaction, so safety and shutdown protocols should be developed for this reactor.

This design aims to maximize the Net Present Value (NPV) by optimization of the key variables: reactor conversion, methanol-to-oxygen inlet molar ratio, carbon monoxide-to-oxygen molar ratio, distillation parameters, and the heat integration network. The reactor is operated at a temperature of 80°C and a pressure of 27 bar, with a conversion of 97% and selectivity of 51%. Our FEL-1 analysis determined that this design has potential profitability, resulting in an NPV of $8 MM and IRR of 19%. This design also has a TCI of $44 MM. To produce 1 kg of dimethyl carbonate, 0.01 kg CO_2 are produced, and 0.02 GJ are used.

Ultimately, we recommend that BICC, Inc moves forward to a FEL-2 design of this plant due to favorable profitability metrics.

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Section 1: Introduction

1.1: Background & Market Analysis

Dimethyl carbonate (DMC) is an organic chemical manufactured globally that has found use as a methylating agent, a green reagent, and most relevantly as a precursor to many polycarbonate resins[1]. The chemical itself is used across many different industries in products such as lithium-ion batteries, adhesives, fuel additives, and in biodiesel production. The dimethyl carbonate market size was valued at $1.1 billion USD in 2023 and is projected to grow to $1.9 billion USD by 2028[2].

As the final polycarbonate product from this site will be used for automobile manufacturing in Mexico, it is highly advantageous to manufacture dimethyl carbonate domestically. Under the USMCA agreement, U.S origin dimethyl carbonate allows the automobile parts which incorporate it to enter the United States duty-free[3]. Dimethyl carbonate is not produced in Mexico, so manufacturing plants are reliant on foreign sources of the chemical, which are subject to tariffs if not sourced in the United States. For instance, dimethyl carbonate imported from China could result in an additional 25% section 301 tariff on top of a 3.1% general duty[4], which would be financially detrimental for a manufacturing sector that sends 76% of its manufactured vehicles to the United States[5]. Thus, domestic manufacturing of dimethyl carbonate is particularly lucrative in light of recent American economic policy changes.

Currently, BICC's processes to produce dimethyl carbonate are reliant on the use of phosgene, which has a highly toxic threshold limit (100 ppb). Finding an alternative production method for dimethyl carbonate that does not involve phosgene and is considered to be less detrimental to the environment is the target of this report.

Our report evaluates the prospective profitability of a oxidative carbonylation process to a chemical plant in order to produce 50 kta of dimethyl carbonate product. Through ASPEN HYSYS and MATLAB simulations, we created a FEL-1 plant design, optimized key design variables including temperature, pressure, reactor conversion, and the inlet molar ratios to determine the Net Present Value (NPV) and Internal Revenue Rate (IRR) of this potential process.

1.2: Reaction Chemistry

Dimethyl carbonate is produced via the exothermic oxidation of carbon monoxide with oxygen and methanol. The process takes place in the liquid phases in the presence of a heterogeneous catalyst and requires moderate temperatures and high pressure. The catalyst needed for this reaction is cuprous chloride (CuCl), a white powder. An additional reaction that takes place is the formation of carbon dioxide via a reaction between carbon monoxide and oxygen.

$$\begin{aligned} &\text{Primary Reaction 1: } 2\mathrm{CH_3OH} + \mathrm{CO} + \frac{1}{2}\mathrm{O_2} \longrightarrow \mathrm{C_3H_6O_3} + \mathrm{H_2O} \quad \Delta H_1^\circ = -300.45 \text{ kJ/mol} \\ &\text{Primary Reaction 2: } \qquad \mathrm{CO} + \frac{1}{2}\mathrm{O_2} \longrightarrow \mathrm{CO_2} \qquad\qquad\quad \Delta H_2^\circ = -284.5 \text{ kJ/mol} \end{aligned} \tag{1.1}$$

The allowable operating temperature for this process is 80 $^\circ$C, with the potential of uncontrollable reaction rates past this temperature as evidenced by our R&D department evaluation. A pressure range between 25 to 30 bar is required for the kinetics in Appendix A to be valid. The primary reaction produces a co-product of water which has a significant deleterious effect on the reaction kinetics. In order to avoid the creation of ineffective catalysts, water must be kept at or below 15 mol% in the reaction liquid by using a molar excess of methanol, greater than 9. The oxygen in both reactions must be the limiting reactant and

cannot be above 4 mol% at any point in the process. This is achieved by setting the carbon monoxide to oxygen molar ratio to be 24 or greater. For this chemistry, there must exist a coupled relationship between the conversion, X, of the limiting reactant oxygen and that of the selectivity, S, of dimethyl carbonate over the limiting reactant oxygen gas. Figure 1.1 represents this relationship as the yield of the dimethyl carbonate product increasing with conversion, but at the expense of selectivity.

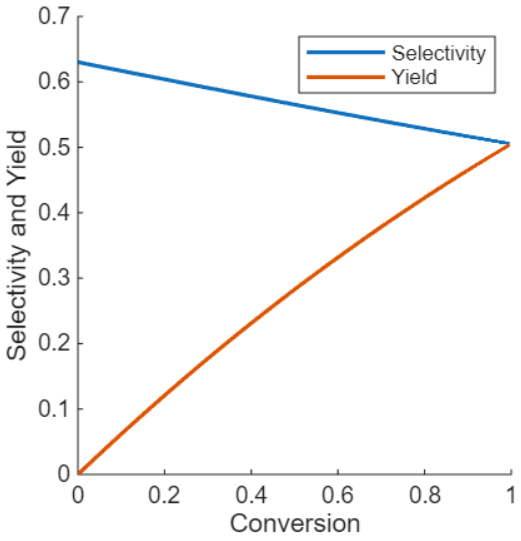


Figure 1.1 : Selectivity as a function of conversion for our dimethyl carbonate chemistry as determined by MATLAB operating at 80˚C, 27 bar, MR_{MeOH} of 9, and MR_{CO} of 24. Only profitable conversions for the dimethyl carbonate process are shown. The optimized conversion is 97% with a corresponding selectivity of 51%.

Section 2: Conceptual Design & Process Overview

Our plant design follows a standard FEL-1 (Front-End Loading) analysis[6] conducted with BICC's methodology.

First, we established foundational data: balanced chemical reactions, target production rates and purities, and process constraints. We also assessed raw material costs, properties, and plant utility requirements to estimate economic feasibility.

Second, we performed an overall mole balance to determine plant input/output flow rates and calculated the desired dimethyl carbonate production based on selectivity. These values were used to estimate the plant's maximum economic potential (Appendix A.1).

Third, we designed the reaction and recycle system. Using material balances and reaction mechanisms, we related reactor conversion, recycle flow, and selectivity to operating parameters—temperature, pressure, carbon monoxide to oxygen molar ratio, and methanol to oxygen molar ratio. We evaluated a range of conditions to determine the required reactor volume and heat duty, then used cost correlations to estimate the capital and operating costs for the reactor and heater.

Fourth, we built the separation system based on the reactor's outlet composition and literature data. A flash drum removes carbon monoxide, carbon dioxide, and oxygen gas for recycling or purge, and remaining separations were arranged by relative volatility. We used ASPEN Plus to estimate distillation stages and column sizing.

Last,we developed a heat exchanger network (HEN) using a comprehensive stream table to reduce previously identified utility costs. We then used MATLAB to optimize reactor conversion, temperature, pressure, and feed ratio for maximum NPV. These values were input into a HYSYS simulation to determine detailed equipment sizes and finalize the process economic analysis.

2.1: Conceptual Design vs HYSYS Simulation

The complete process analysis given by the HYSYS simulation is slightly different than that given by the MATLAB conceptual design (Appendix E). These values differ due to the assumptions of each software system. Those include:

1. MATLAB treats all gasses as ideal and assumes that the heats of reaction are constant. Both systems use the Aly & Lee, or DIPPR thermodynamic equation to calculate the heat capacity. The HYSYS simulation also uses the N thermodynamic package and is able to more accurately calculate and model thermodynamic behavior. For our HYSYS environment, we used a combination of Peng Robinson (PR) for the high pressure reactor environment and Non-Random Two Liquid (NRTL) for the column environments for lower pressures (< 10 bar).
2. The mole balances for the flash were modeled after a perfect flash and feature complete separation of methanol, dimethyl carbonate, and water split from the gases of carbon monoxide, oxygen, and carbon dioxide. This assumption alters the amount of dimethyl carbonate theoretically recovered, but our designed flash conditions approximate this assumption quite well with less than 1% of the dimethyl carbonate feed lost to the vapor. Recovery of said dimethyl carbonate is provided by the scrubber unit before the recycle.
3. We make the assumption that the fresh feed of methanol is pure, but it will typically contain trace amounts of other hydrocarbons such as ethylene or propylene. This assumption allowed the methanol coming from the separator to be recycled back without the need for a purge. In reality, due to the trace components coming in with the methanol, a purge would be required to ensure there is no buildup of the trace components[7].

2.2: Process Flow Diagram

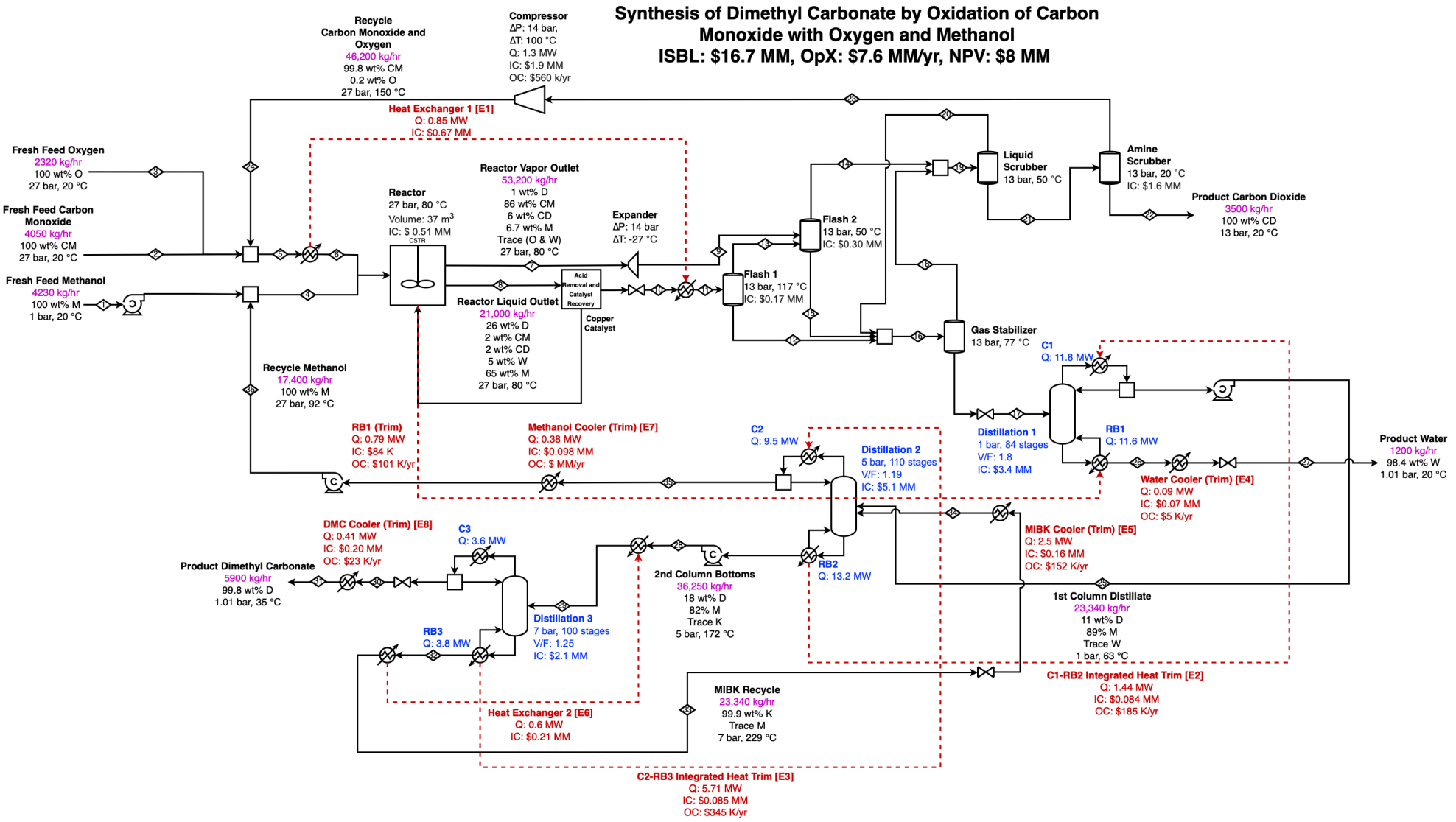


Figure 2.2 : Process flow diagram of the optimized DMC plant. For a detailed overview of the mass flow rates, compositions, temperature and pressure of each stream, see the stream summary in Table 2.1.

Table 2.1 : Component streams as modeled on HYSYS under the assumption that the process operates on plant years (8400/yr)

Stream	Name	Temperature (°C)	Pressure (bar)	Mole Composition	Mass Flow Rate (kg/hr)	Mass Flow Rate (kta)
1	Methanol Feed	25.00	1.0	100% M	4,235	35.5
2	Carbon Monoxide Feed	20.00	27.0	100% CM	4,070	34.2
3	Oxygen Feed	20.00	27.0	100% O	2,325	19.5
4	Combined Liquid	80.00	27.0	100% M	21,650	182
5	Combined Gas	134.60	27.0	96% CM, 4% O	18,400	442
6	Reactor Gas Feed	80.00	27.0	96% CM, 4% O	18,400	442
7	Reactor Liquid Effluent	80.00	27.0	10.66% D, 2.43% CM, 1.47% CD, 10.49% H, 74.95% M	21,040	177
8	Reactor Vapor Effluent	80.00	27.00	0.12% O, 0.37% D, 89.26% CM, 3.90% CD, 0.31% H, 6.05% M	53,180	447
9	Decompressed Vapor	53.00	13.00	0.12% O, 0.37% D, 89.26% CM, 3.90% CD, 0.31% H, 6.05% M	53,180	447
10	Decompressed Liquid	78.25	13.0	10.66% D, 2.43% CM, 1.47% CD, 10.49% H, 74.95% M	21,040	177
11	Liquid Flash Inlet	117.45	13.0	10.66% D, 2.43% CM, 1.47% CD, 10.49% H, 74.95% M	21,040	177
12	Liquid Flash Bottom	117.45	13.0	11.02% D, 0.41% CM, 1.18% CD, 10.83% W, 76.55% M	20,030	171
13	Liquid Flash Tops	117.45	13.00	0.11% O, 2.10% D, 50.16% CM, 8.18% CD, 2.31% W, 37.13% M	739	6.23
14	Gas Flash Tops	50.25	13.0	0.12% O, 0.16% D, 91.60% CM, 4.04% CD, 0.11% W, 3.97% M	51,850	436
15	Gas Flash Bottoms	50.25	13.0	0.01% O, 7.51% D, 2.04% CM, 1.12% CD, 7.18% W, 82.14% M	2,072	17.4
16	Scrubber Feed	105.35	13.0	9.90% D, 0.51% MO, 1.05% CD, 9.61% W, 78.93% M	24,950	210

Stream	Name	Temperature (°C)	Pressure (bar)	Mole Composition	Mass Flow Rate (kg/hr)	Mass Flow Rate (kta)
17	1st Column Feed	77.05	2.00	10.06% D, 9.76% W, 80.18% M	24,540	206
18	Scrubber Recycle	76.85	13.0	0.12% O, 32.61% CM, 67.28% CD	411	3.5
19	Liquid Scrubber Inlet	50.45	13.0	0.12% O, 0.16% D, 91.25% CM, 4.42% CD, 0.11% W, 3.95% M	52,260	100
20	Scrubbed Liquids	50.25	13.0	3.7% D, 2.66% W, 93.64% M	2,571	22
21	Amine Scrubber Inlet	20.00	13.0	100% CD	3,507	29.5
22	Carbon Dioxide Production	20.00	13.0	100% CD	3,507	29.5
23	Gas Recycle	50.25	27.0	99.87% CO, 0.13% O	46,180	388
24	Compressed Gas Recycle	50.25	27.0	99.87% CM, 0.13% O	46,180	388
25	1st Column Distillate	63.15	1.0	11.16% D, 0.04% W, 88.81% M	23,340	196
26	1st Column Bottoms	98.05	1.0	98.39% W, 1.61% M	1,197	16.5
27	Water Production	25.00	1.0	100% W	1,197	16.5
28	2nd Column Bottoms	171.5	5.0	17.79% D, 0.08% M, 82.14 K	36,250	305
29	3rd Column Inlet	188.85	5.0	17.79% D, 0.08% M, 82.14 K	36,250	305
30	3rd Column Distillate	164.85	7.0	99.8% D, 0.15% M, 0.07% K	5,896	50
31	DMC Production	25.00	1.0	99.8% D, 0.15% M, 0.07% K	5,896	50
32	3rd Column Bottoms	228.5	34.0	86.57% M, 13.43% H	9,050	76
33	MIBK Recycle	201.15	7.0	0.09% D, 99.91% K	30,360	255
34	MIBK Feed	228.5	5.0	100% K	30,400	255
35	Methanol Recovery	111.55	5.0	0.38% D, 99,62% M	17,420	146
36	Methanol Recycle	91.75	27.0	100% M	17,420	146

2.3: Optimization of Key Design Variables

The initial optimization of the dimethyl carbonate process required external simulation through MATLAB algorithm featuring reactor and mole balance calculations. Key design variables are $\mathrm{MR}_{\mathrm{CO}}$, the molar ratio of carbon monoxide to oxygen in the feed to the reactor, and $\mathrm{MR}_{\mathrm{MeOH}}$, the molar ratio of methanol to oxygen in the feed to reactor. These variables were varied to find selectivity, conversion, and liquid reactor residence time to obtain the maximum NPV.

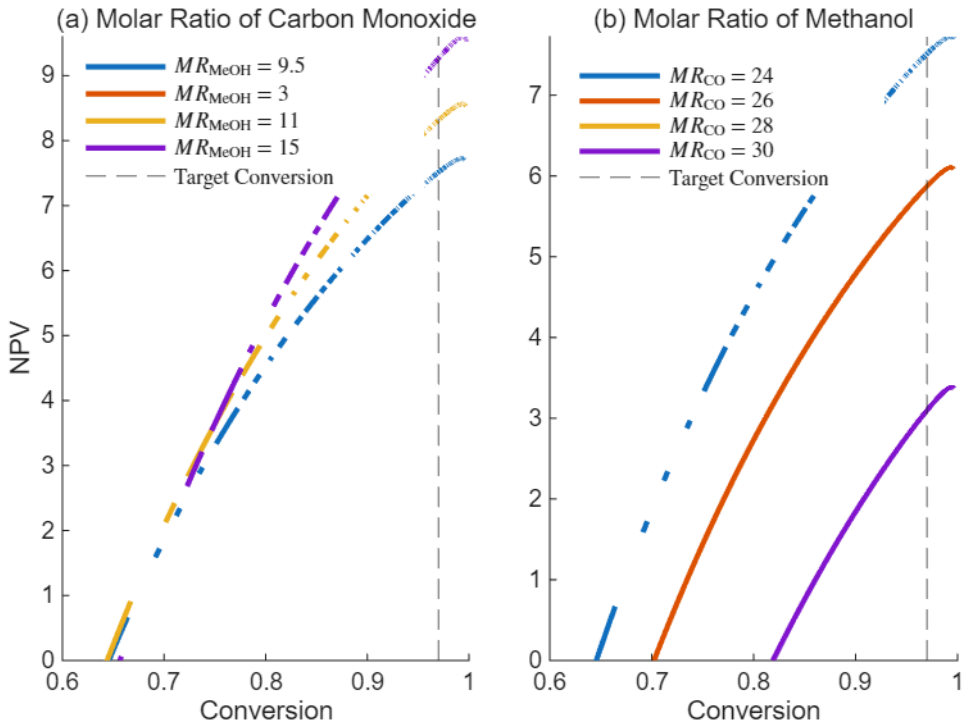


Figure 2.3 : NPV as a function of conversion for select values of molar ratio of carbon monoxide and methanol. Note that the NPV climbs higher despite our desired case being lower conversion of 97%, this was chosen due to achievable values in HYSYS.

Figure 2.3 represents the NVP as a function of conversion for multiple molar ratios of methanol and carbon monoxide to oxygen. The gaps in the lines depict invalid cases at the specified molar ratio.

However, the MATLAB algorithm uses a minimum energy of separation, W_{min}, as the basis for modeling the ISBL and OPEX costs for the separation subsystem, which does not account for heat integration potential. Too little heat integration for the separation system often made any plant design unprofitable. To work around this, the decision was to alter our process design parameters achieve a desirable heat integration.

Hence, graphs shown in Figure 2.3 are not representative of actual NPV values achieved on HYSYS, rather they are some constant multiple of the real NPV calculated though HYSYS.

2.4: Reactor Design

Reactor volume, V, is impacted primarily through conversion of methanol. The production of dimethyl carbonate requires the reaction to occur in the liquid phase with pressurized gas allowed to dissolved into the liquid phase with a residence time, τ. Figure 2.4 shows that conversion increases with residence time due to the carbon monoxide and oxygen dissolving into the liquid phase.

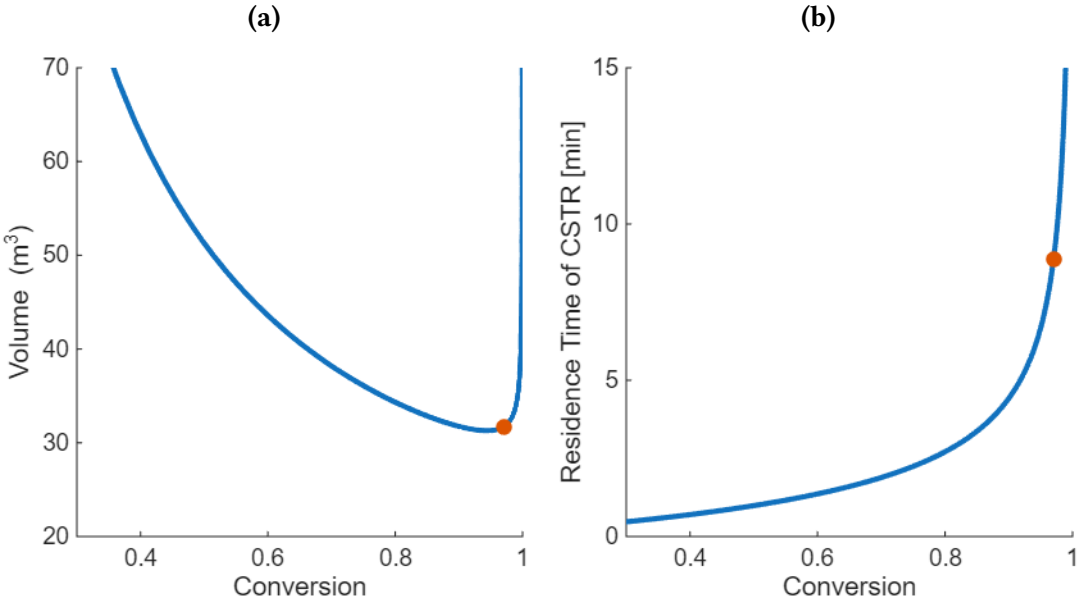


Figure 2.4 : Conversion vs residence time and volume of CSTR of for the isothermal reaction kinetics model. The red dot indicates the optimal conditions predicted by the MATLAB conceptual design with X = 0.97 and $\tau = 9$ min.

2.5: Separation System Design

The HYSYS design for all three distillation columns makes a constant molal overflow assumption so the molar flow rates of the liquid and vapor are constant throughout the columns.

Two azeotropes were found between the components of this design. There is an azeotrope between dimethyl carbonate and water, as well as between dimethyl carbonate and methanol. The separation of the ternary group dimethyl carbonate, water, and methanol was performed using 3 distillation columns in series, first separating water as the heavy key from the 2 lighter components, and then separating methanol from dimethyl carbonate using methyl isobutyl ketone. This entrainer was selected due favorable separation abilities for dimethyl carbonate and methanol as well as reduced process carbon dioxide emissions as outlined by Varyemez and Kaymak[8]. The system follows a design outlined in Matsuda et al.[9], which followed the procedure in Doherty and Malone[10] to determine that the optimal MIBK to feed ratio was 0.56. Since the feed composition of our system and the system found in literature were similar, the literature values for optimized stages and MIBK to feed ratio were used to achieve the desired purity of dimethyl carbonate. The distillation parameters used in this system were optimized from these values to produce effective and precise component separation.

The first column has 42 theoretical stages, a vapor to feed molar flow rate (V/F) ratio of 1.8, and a specified 0.9999 recovery of dimethyl carbonate in the distillate. The column is operated at 1 bar assuming no pressure drop along the column. The second column has 55 theoretical stages, a V/F ratio of 1.19, and a specified 0.9995 recovery of methanol in the distillate, operated at 5 bar. Lastly, the final column has 50 theoretical stages, a V/F ratio of 1.25, and a specified 99.8 wt.% of dimethyl carbonate in the distillate, operated at 7 bar. The latter 2 columns are operated at greater than atmospheric pressure as it keeps outlet feeds at a reasonable temperature.

2.6: Heat Exchanger Network Design

This oxidative carbonylation process is energy intensive and requires a robust heat integration network to reduce operating costs for improved profitability. It is assumed that a start-up furnace will need to be

included in plant operation, and is accounted for in start-up costs for FEL-1 considerations. First, 2 heat exchangers were proposed between process streams to remove the need for heating and cooling utilities within that part of the plant, which are detailed further in Appendix D. The inclusion of these process heat exchangers reduce operating energy by a combined total of 1.32 MW, lowering capital and operating costs. Significant energy expenditures stem from the distillation column system, so heat integration of the column reboilers and condensers working fluids was implemented for reduced operating costs. To ensure adequate column heating and cooling, heat exchangers were designed (Appendix D) to allow the condenser and reboiler sets to share utilities while remaining within allowable temperature and material design limits. Installed costs for this system's heat exchangers were incorporated into the installed cost of the distillation column. This results in a staggering energy requirement decrease from 57.1 MW to 10.8 MW, a 81.2% duty reduction through the heat integration network.

To additionally lower energy usage, a working fluid loop was designed to use energy removed from the reactor to keep the unit isothermal to then heat the reboiler on the 1st distillation column. The loop design is shown in Appendix D and is modeled similarly to the working loop between the reboilers and condensers within the separation system. This allows for these units to share utilities, lowering the working fluid necessary to produce desired temperature changes in both the reactor and reboiler. This design results in a 96% reduction in total duty (22.34 MW to 0.79 MW), and its heat exchanger installation costs are included in the reactor installed cost calculation.

As this are preliminary designs, we recommend FEL-2 conduct a more detailed analysis to refine the installed cost estimations and operational requirements.

Section 3: Economic Analysis

3.1: Economic Metrics & Analysis

The final process economic analysis (Table 3.2) was calculated using the HYSYS simulation feed inlets, product outlets, heater duty and reactor volume at the ideal conditions determined through MATLAB and HYSYS optimization. For more favorable profitability margins via reduced energy expenditures, some parameters were first calculated in MATLAB and then further optimized in HYSYS.

Equipment included in installation costs for this plant are the compressor, reactor, separation system, amine scrubber and heat exchanger network. The mixers, pumps, and expander are not included in these economic calculations because they are beyond the scope of this report. The land cost for this plant is also not included under the assumption that overall plant size and exact location are unknown. The total ISBL cost for this process is $16.7 MM and the OSBL cost is $6.7 MM. The direct cost is $9.7 MM and the indirect cost is $7.0 MM. These values bring the fixed capital investment to a total of $38 MM when including a 25% contingency factor to ensure all unexpected costs during construction can be covered. The working capital value is $3.6 MM and the start-up cost is $2.2 MM, bringing the total capital investment to $44 MM.

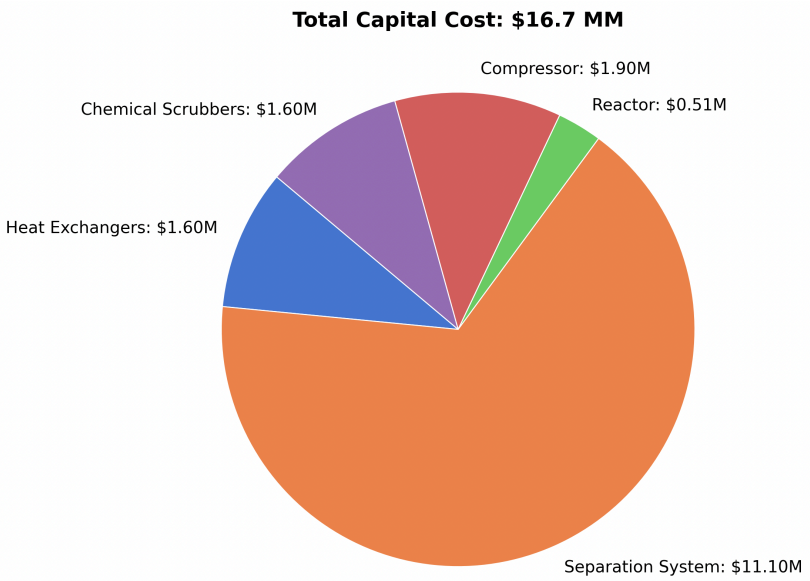


Figure 3.5 : Pie chart of the capital costs for the plant design proposal.

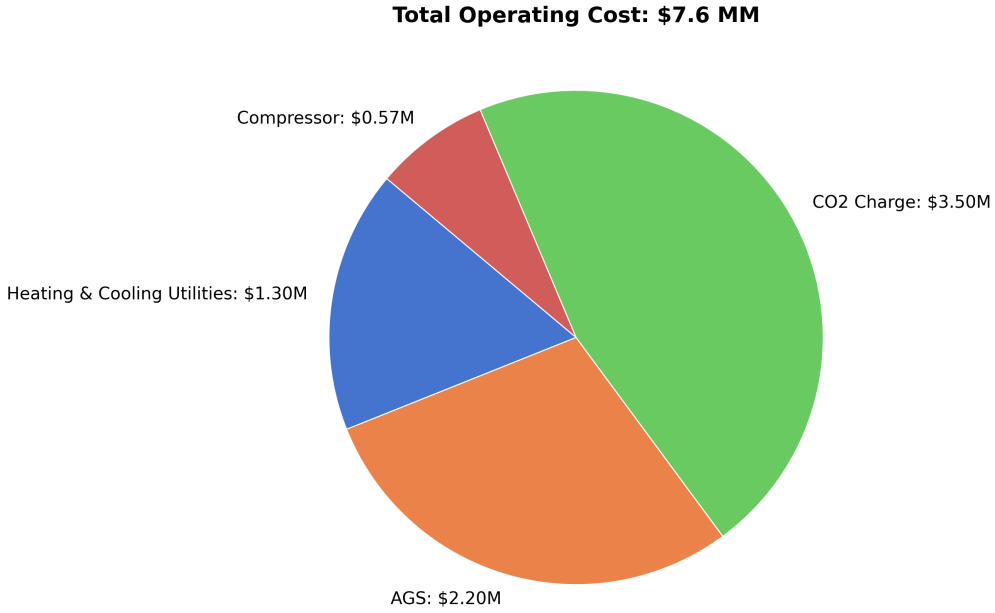


Figure 3.6 : Pie chart of the operating costs for the plant design proposal.

Table 3.2 : Final economic calculations determined using HYSYS parameters.

Economic Metric	Value
NPV ($MM)	8
IRR (%)	19
TCI ($MM)	44
OPEX ($MM/yr)	8

The operating costs for this plant include utilities used to power the process units, administrative & general services (AGS), carbon production charge, and wastewater treatment (See Appendix D.3 for cost

correlations). The cost of the inlet feed is $21.8 MM. The total cost of operational utilities is $1.3 MM. The administrative and general service costs total to $2.2 MM and include administration costs, product selling prices, and plant research & development, assuming these values include salary, overhead, maintenance costs, and financing interest. The carbon charge fee is $100/MT and costs $3.5 MM. We assume that the Scope 1 emissions are equal to the amount of carbon dioxide produced in the undesired side reaction. Scope 2 emissions, which consist of emissions produced from units that use electricity or natural gas are included in the overall carbon charge. This brings the overall plant operating cost to $7.6 MM.

3.2: Reactor Costs

The installation cost of the continuous stirred reactor is based on the assumption that it is a jacketed reactor and is composed of teflon-lined carbon steel, which provides resistance against hydrochloric acid corrosion caused by catalyst consumption. The reactor is $505K to install, and its operating cost is factored into the heat exchanger it shares with the reboiler on the first distillation column, where a trim heater is added to account for the duty difference between the units.

3.3: Compressor Costs

The compressor is assumed to be a motor drive centrifugal compressor and has an installed cost of $1.1 MM. The operating cost is based on the operating power of the compressor and assumes that the compressor is electricity powered. This sets the operating cost as a function of the operating power and the price of electricity in kWh, valued at $0.05/kWh, to be $559K.

3.4: Separation System Costs

The plant separation system consists of 2 flash drums, 3 distillation columns, a liquid scrubber, an amine scrubber and a stabilizer. The process units within this system were built with 316 stainless steel to ensure resistance against corrosion. The installed cost of the flash drums are calculated using the HYSYS simulation as they are not part of the MATLAB model due to its assumptions. The first column contains 42 theoretical stages with a 99.99% fractional recovery of dimethyl carbonate in the distillate and an installed cost of $3.5M. The second distillation column has 55 theoretical stages with a 99.9% fractional recovery of methanol in the distillate and an installed cost of $5.1M. The third distillation column has 50 stages with 99.8 wt.% DMC in the distillate and an installed cost of $2.2M. The amine scrubber installed cost was calculated using $\dot{W}_{\text{min}}$, the minimum work, λ, the estimated ratio of the true work to the minimum work, and $C_{\$/\text{W}}$, the capital cost correlation factor and is $1.8M. The flash drums, liquid scrubber, and stabilizer are not costed in this system as they are part of FEL-2 financial considerations.

3.5: Sensitivity Analysis

To produce viable economic analysis based on our FEL-1 plant design, we must rely on multiple assumptions regarding various economic factors including dimethyl carbonate, methanol, and oxygen prices holding constant with time, as well as the tax and enterprise rates staying at their current rates of 27% and 15% respectively. To evaluate the sensitivity of the net present value to these economic fluctuations, we calculated the range of changes possible based on expected variations in different factors.

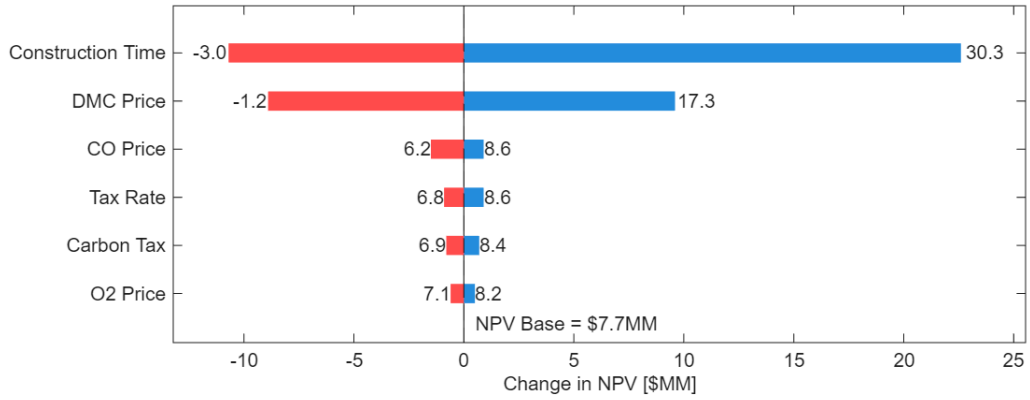


Figure 3.7 : Variations in NPV as a result of fluctuations in key economic variables.

The factor with thee most influence over NPV is the construction time. The NPV when the construction time is altered has the greatest fluctuations, in a range of $–3.0MM to $30.3MM. This range of sensitivity is an indicator that implementing this process requires strict adherence to the construction schedule, or plant profitability could greatly suffer. Extensions to construction time risks the plant's ability to be financially positive. Furthermore, we must also consider the economic impact of variations in dimethyl carbonate prices, which could result in a profitability range of $–1.2MM to $17.3MM, a staggering change in financial returns. Based on recent economic policies enacted by the United States, this price is likely to change over the next year. The United States is the 2nd largest importer of dimethyl carbonate[11] and has set intense tariffs on top exporters Vietnam, China, and South Korea[4,12]. If these tariffs remain after plant operation begins, it will be far more lucrative to produce dimethyl carbonate domestically to support American importers.

In addition, it was determined that the plant would generate an NPV of zero at the end of its life if the price of methanol increases by 26%, carbon monoxide by 72%, and oxygen by 290%, respectively. The NPV also drops to zero if the selling price of dimethyl carbonate decreases by 8.8%. From these findings, it appears that American economic policy decisions will be integral in evaluating the continued viability of this plant design.

3.6: Second-Order Effects: Monte Carlo Simulation

To further quantify the risks involved in this plant's design with respect to the economic viability, we applied a Monte Carlo simulation to measure the financial risk associated with abnormal economic situations, including a national or global recession. In our simulation, we applied a triangular distribution to model the underlying distribution of three factors: the prices of dimethyl carbonate, raw materials, and the construction time of the project.

We conclude that this design contains a 10% probability of this project failing to return profit. In general, we have a 52% chance of obtaining a NPV less than our base case of $8 MM and a 48% chance of obtaining a higher NPV.

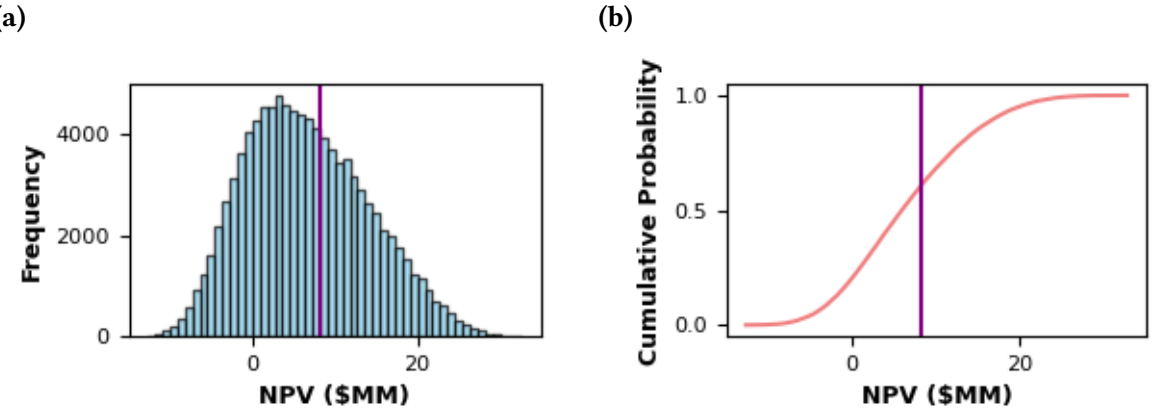


Figure 3.8 : **(a)** Distribution of our NPV after applying Monte Carlo simulation featuring changes construction time and prices of feeds **(b)** The calculated probability distribution which measures the probability of achieving a certain NPV given the applied perturbations to our design.

Triangular distributions were employed for all variables as our goal was to measure deviations around the most frequent or common value (mode-based distribution). In addition, the population distribution for most variables are often unknown and the triangular distribution provides a well-known simple and effective solution to modeling said parameters.

Section 4: Safety and Environmental Impact

4.1: Safety Hazards

There are important safety considerations necessary for plant function, involving the operating conditions, chemical components, and reactions within this process. A comprehensive HAZOP analysis (Appendix E) identified that the greatest threat to both human life and plant integrity lies in the carbon monoxide and oxygen mixture. A mixing unit combines fresh and recycled carbon monoxide and oxygen. Carbon monoxide and oxygen combinations are highly explosive and can lead to catastrophic failure; any blockage or failure in the carbon dioxide feed line can cause the mixture to exceed the oxygen lower limit of explosion of 4 mol%. A small delay in detecting a carbon monoxide blockage can cause an accumulation of oxygen in the mixture stream. To mitigate the risk of oxygen exceeding the lower limit of explosion, the process uses fail-open inert gas valves. Even without an explosion, a carbon monoxide leak would endanger workers. Carbon monoxide is a colorless and odorless gas that replaces oxygen in red blood cells, causing hypoxia and hundreds of preventable deaths annually[13].

The exothermic reaction becomes hazardous in the CSTR if cooling fails. The temperature rises significantly if the reaction is not operated under isothermal conditions. At higher temperatures, there is an acceleration in reaction rates, causing the reaction to become uncontrollable, decomposition of dimethyl carbonate[14], and pressure escalations. To prevent vessel rupture, the reactor runs isothermally using fail-open quench valves for rapid reaction suppression and fail-close feed valves to isolate the reactants, ensuring any temperature rise immediately halts the process. During FEL-2 considerations a backup cooling system should be designed to ensure safe reactor operation.

Dimethyl carbonate storage and transfer are flammable and pose flash fire risks. An incident in 2013 resulted in the death of an operator when dimethyl carbonate vapors ignited during barrel tipping. As a result, closed-cycle, grounded, nitrogen-blanketed systems are recommended[15]. In addition, level sensors inside the storage tank are set with both high-level and low-level alarms. On high level, the inlet will fail-

close to prevent overfill, and on low level, the output pump will fail-close to prevent dry running and static buildup.

4.2: Environmental Impact

To evaluate the extent of this chemical plant's environment impact, it is important for us to consider the carbon dioxide being produced and then released. In total, 621K GJ are used for an overall production of 35K MT annually, where 0.01 kg CO_2 is produced for every 1 kg of $C_3H_6O_3$. These consider the CO_2 produced by the process as well as the energy and steam used for the heating utilities and compressor that indirectly produce CO_2.

Table 4.3 : Environmental hazards that product streams may contribute to the environment.

Component	Environmental Hazard
Methanol $[CH_3OH]$	Methanol may have an effect on animals, birds, and fish. In some cases it can lead to their death. Exposure can also cause low growth rate in plants.[16]
Carbon Monoxide $[CO]$	Carbon monoxide effects the amount of greenhouse gases, which are linked to climate change. This results in land and sea temperatures increases and a change to ecosystems.[17]
Oxygen $[O_2]$	Excess amounts of oxygen can cause organ damage, seizures, and even death in animals. At high levels, oxygen is toxic to every life form.[18]
Dimethyl Carbonate $[C_3H_6O_3]$	Dimethyl carbonate has a low environmental toxicity. It is not a hazard to aquatic species.[19]
Carbon Dioxide $[CO_2]$	Once carbon dioxide enters the atmosphere it warms up the planet, causing climate change.
Cuprous Chloride $[ClCu]$	Cuprous chloride is highly toxic to aquatic life with lasting effects. Spillage can cause pollution of water courses.[20]
Water $[H_2O]$	No environmental hazard
Hydrochloric Acid $[HCl]$	Hydrochloric acid can affect the the acidity in water and affect aquatic organisms.[21]
Methyl-Isobutyl-Ketone $[C_6H_{120}]$	Methyl-Isobutyl-Ketone has a slight acute toxicity to aquatic life and high toxicity to birds.[22]

The product streams may pose additional environmental hazards beyond carbon dioxide release, detailed in Table 4.3. Wastewater treatment of water exiting the plant is outlined in Appendix D.10, but is very minor in the scope of the plant. In summary, most of our products are short-term environmental hazards that can cause acute harm, with carbon monoxide and carbon dioxide being intense greenhouse gases.

Section 5: Process Alternatives and FEL-2 Considerations

The main process alternative considered was operating the plant with a purge stream, where instead of completely removing CO_2 from the recycle gas mixture, a part of that gas mixture would be purged and recycled back to the process with CO_2. However, it resulted in a lower NPV, as it is more cost-intensive in terms of purchasing reactants. Purging the gas mixture containing O_2, CO, and CO_2 results in the loss of reactants, forcing the cost of fresh feed to increase.

The main process alternatives considered for the separation system were pressure swing distillation and utilizing a decanter along with liquid-liquid extraction. Pressure swing distillation would require one

distillation column to operate at relatively low pressures and a second column to operate at high pressures. However, this process resulted in a lower NPV due to the higher capital and operating costs associated with the high-pressure column. The decanter, along with liquid-liquid extraction, would require methanol to be separated from water and dimethyl carbonate before adding the decanter and performing liquid-liquid extraction. This process also resulted in a lower NPV.

To maximize dimethyl carbonate production, an acid removal system must be implemented. A potential unit that is capable of removing hydrochloric acid (HCl) in compliance with the Indiana Department of Environmental Management (IDEM) minimum acid removal limits is a neutralization tank. The tank will combine a basic compound (sodium hydroxide) with corrosive HCl to neutralize the acid, which can then be removed from the reactor. An individual air permit will need to be awarded by IDEM for the plant to be operational[23]. Cost estimates should be prepared during FEL-2 evaluation of this process design.

Section 6: Conclusion

Our FEL-1 Analysis demonstrated that the proposed DMC process has the potential to be profitable at a production rate of 50 kta of dimethyl carbonate. The NPV of the plant is $8 MM and the IRR is 19%. The TCI of the plant is $44 MM. Based on our sensitivity analysis, it is likely these calculations will fluctuate with market pricing and economic cycles in the commodity chemical industry.

The 37 m^3 continuous stirred tank reactor for this plant operates at 80°C and 27 bar to produce 50 kta of dimethyl carbonate. The reactor conversion is 97%, and unreacted methanol, carbon monoxide, and water are separated and recycled back to the reactor feed. To prevent hazardous operation of this process, safety protocols and standards must be implemented.

Ultimately, we recommend that BICC, Inc. moves forward to a FEL-2 design of this plant due to favorable profitability metrics.

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Appendix A: Reactor Design and Mole Balances

A.1: Overall Plant Balances and Degree of Freedom Analysis

$$\begin{aligned} &\text{Primary Reaction 1: } 2\mathrm{CH_3OH} + \mathrm{CO} + \frac{1}{2}\mathrm{O_2} \longrightarrow \mathrm{C_3H_6O_3} + \mathrm{H_2O} \\ &\text{Primary Reaction 2: } \qquad \mathrm{CO} + \frac{1}{2}\mathrm{O_2} \longrightarrow \mathrm{CO_2} \end{aligned} \tag{A.1}$$

We also have the follow shorthand symbols used throughout this appendix to describe all balances:

$$M \equiv \mathrm{CH_3OH}, \mathrm{D} \equiv \mathrm{C_3H_6O_3}, \mathrm{CM} \equiv \mathrm{CO}, \mathrm{CD} \equiv \mathrm{CO_2}, \mathrm{W} \equiv \mathrm{H_2O}, \mathrm{O} \equiv \mathrm{O_2} K \equiv \mathrm{C_6H_{12}O} \tag{A.2}$$

A material balance on each of the components involved in these reactions produces the following system of equations, where ξ_i is defined as the extent of the ith reaction. Using the steady-state process balance *in - out + gen - cons = 0*, the following mole balances are produce

$$\begin{aligned} M &: F_M = 2\xi_1 \\ \mathrm{DMC} &: P_\mathrm{D} = \xi_1 \\ \mathrm{CM} &: F_\mathrm{CM} = \xi_1 + \xi_2 \\ \mathrm{CD} &: P_\mathrm{CD} = \xi_2 \\ W &: P_W = \xi_1 \\ O &: F_O = \frac{1}{2}\xi_1 + \frac{1}{2}\xi_2 \end{aligned} \tag{A.3}$$

where F_i is the molar flow rate of component i into the plant and P_i is the molar flow rate of component i out of the plant. For 2 reactions, we require 2 reference compounds, dimethyl carbonate and carbon dioxide. Hence the remaining balances of methanol, oxygen, carbon monoxide, and water can be rearranged to determine the global input-output balances (IO Level):

$$\begin{aligned} F_M &= 2P_\mathrm{D} \\ F_O &= \frac{P_\mathrm{D}}{2S} \\ F_\mathrm{CM} &= \frac{P_\mathrm{D}}{S} \\ P_W &= P_\mathrm{D} \end{aligned} \tag{A.4}$$

In addition, S is defined as the selectivity of dimethyl carbonate with respect to oxygen.

$$S = \frac{P_\mathrm{D}}{F_O} \tag{A.5}$$

Rewriting the mole balances in Equation (A.4) with Equation (A.5) above and reducing to just design variables (S and P_{DMC}) gives the following IO Level balances

$$\begin{aligned} F_M &= 2P_{\mathrm{D}} \\ F_O &= \frac{P_{\mathrm{D}}}{2S} \\ F_{\mathrm{CM}} &= \frac{P_{\mathrm{D}}}{S} \\ P_{\mathrm{CD}} &= -\frac{P_{\mathrm{D}}(-1+S)}{S} \\ P_W &= P_{\mathrm{D}} \end{aligned} \tag{A.6}$$

A.2: Recycle and Reactor Mole Balances

Let $E_{\mathrm{i,liq}}$ be the liquid effluent of specie i out of the reactor, $E_{\mathrm{i,vap}}$ be the vapor effluent of specie i out of the reactor, I_i be the feed into the reactor, and R_i to be the recycle flow rate of specie i.

The reactor per-pass efficiency X can be defined using the relationship below:

$$\mathrm{X} = \frac{\frac{1}{2}\xi_1 + \frac{1}{2}\xi_2}{2I_O} = \frac{P_{\mathrm{D}} + P_{\mathrm{CD}}}{2\big(P_O + R_O - \frac{1}{2}(P_{\mathrm{D}} + P_{\mathrm{CD}})\big)} \tag{A.7}$$

The other design variables of interest are the molar ratio of carbon monoxide to oxygen into the reactor and the molar ratio of methanol to oxygen into the reactor:

$$\begin{aligned} \mathrm{MR}_{\mathrm{CM}} &= \frac{P_{\mathrm{CM}} + R_{\mathrm{CM}} + P_{\mathrm{D}}}{P_O + R_O - \frac{1}{2}(P_{\mathrm{D}} + P_{\mathrm{CD}})} \\ \mathrm{MR}_M &= \frac{F_H + R_H}{P_O + R_O - \frac{1}{2}(P_{\mathrm{D}} + P_{\mathrm{CD}})} \end{aligned} \tag{A.8}$$

Finally, we can do a molar balance around the reactor with all outgoing flow rate written in E_i and I_i notation.

$$\begin{aligned} M &: E_{\mathrm{W,vap}} + E_{\mathrm{W,liq}} = I_M - 2\xi_1 \\ \mathrm{D} &: E_{\mathrm{D,vap}} + E_{\mathrm{D,liq}} = \xi_1 \\ \mathrm{CM} &: F_{\mathrm{CM}} = \xi_1 + \xi_2 \\ \mathrm{CD} &: P_{\mathrm{CD}} = \xi_2 \\ W &: P_W = \xi_1 \\ O &: F_O = \frac{1}{2}\xi_1 + \frac{1}{2}\xi_2 \end{aligned} \tag{A.9}$$

Additional, we have the balance around the amine scrubber system and separation system:

$$\begin{aligned} O &: E_{\mathrm{O,vap}} + E_{\mathrm{O,liq}} = R_O \\ \mathrm{CM} &: E_{\mathrm{CM,vap}} + E_{\mathrm{CM,liq}} = R_{\mathrm{CM}} \\ \mathrm{CD} &: E_{\mathrm{CD,vap}} + E_{\mathrm{CD,liq}} = R_{\mathrm{CD}} \end{aligned} \tag{A.10}$$

$$\begin{aligned}
&\text{Separation :}\\
&M : E_{\mathrm{M,vap}} + E_{\mathrm{M,liq}} = R_M\\
&\mathrm{D} : E_{\mathrm{D,vap}} + E_{\mathrm{D,liq}} = R_{\mathrm{D}}\\
&W : E_{\mathrm{W,vap}} + E_{\mathrm{W,liq}} = R_W
\end{aligned} \tag{A.11}$$

Put all together and written in terms of purely design variables ($S, \mathrm{X}, \mathrm{MR}_M, \mathrm{MR}_{\mathrm{CM}}, P_{\mathrm{DMC}}$) we have the following expressions:

$$\begin{aligned}
R_M &= \frac{P_{\mathrm{D}}(\mathrm{MR}_M + 4S\mathrm{X})}{2S\mathrm{X}}\\
R_{\mathrm{CM}} &= \frac{P_{\mathrm{D}}(\mathrm{MR}_{\mathrm{CM}} - 2\mathrm{X})}{2S\mathrm{X}}\\
R_0 &= -\frac{P_{\mathrm{D}}(-1 + \mathrm{X})}{2S\mathrm{X}}\\
I_M &= \frac{\mathrm{MR}_M P_{\mathrm{D}}}{2S\mathrm{X}}\\
I_{\mathrm{CM}} &= \frac{\mathrm{MR}_{\mathrm{CM}} P_{\mathrm{D}}}{2S\mathrm{X}}\\
I_O &= \frac{P_{\mathrm{D}}}{2S\mathrm{X}}
\end{aligned} \tag{A.12}$$

A.3: Derivation of Reactor Design Equations

A.3.1: Reaction System Rate Laws

The reaction rates for chemical reactions 1 & 2 are:

$$r_1 = k_1 C_M^2 C_O^{0.5} \tag{A.13}$$

$$r_2 = k_2 C_O^{0.5} \tag{A.14}$$

with pre-factors $k_{1,\mathrm{f}}$ and $k_{2,\mathrm{f}}$ determined by the expressions

$$k_1 = k_{1,0} \exp\left(-\frac{E_{\mathrm{a1}}}{RT}\right) \tag{A.15}$$

$$k_2 = k_{2,0} \exp\left(-\frac{E_{\mathrm{a2}}}{RT}\right) \tag{A.16}$$

Here, T is in K, the activation energies are in J/gmol, and concentrations are in gmol/L. The rate and rate constant equations are viable for a temperature of 80°C, a pressure range of 25 bar $< P <$ 30 bar, a carbon monoxide to oxygen molar ratio range of 24 or greater, and a methanol to oxygen molar ratio range of 9 or greater.

A.3.2: Reactor Material Equations

In order to find the conversion and volume of the CSTR at desired design variable conditions, we developed the CSTR material equations with our given reaction rates. The component material balance equations take the generic form:

$$N = N_0 - V_R \sum_{j=1}^{R} \nu_{ji} r_j \tag{A.17}$$

where c_i is the component molar concentration, V_R is the reactor volume, and R_i is the production rate for the component. Where $R_i = \Sigma\,(r_i)$, V_R can be divided out of both sides to produce the following ODEs for the change in molar flow rate for each component.

$M : \frac{dC_M}{dt} = -2r_1$

$\mathrm{CM} : \frac{dC_{\mathrm{CM}}}{dt} = -r_1 - r_2$

$\mathrm{CD} : \frac{dC_{\mathrm{CD}}}{dt} = r_2$

$D : \frac{dC_D}{dt} = r_1$

$O : \frac{dC_O}{dt} = -0.5r_1 - 0.5r_2$

A.3.3: Reactor Design

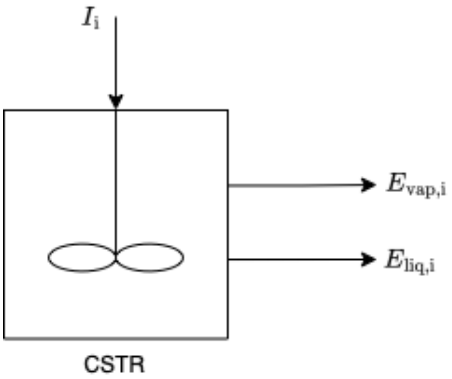


Figure A.9 : Illustration of the reactor with feed flow rate I_i and outlet flow rates $E_{\mathrm{vap},i}$ and $E_{\mathrm{liq},i}$.

The relationship between selectivity and conversion are independent of reactor scale, which allows us to calculate the volume of the reactor and its conversion and selectivity based on a reactor feed rate basis. In this case, the temperature was assumed to be 80˚C with an $\mathrm{MR}_{\frac{M}{O}}$ of 9.2 and an $\mathrm{MR}_{\frac{\mathrm{CM}}{O}}$ of 24. It is assumed that only methanol, carbon monoxide, and carbon dioxide are recycled and are the only components at the feed of the reactor. Assuming steady-state we get the following:

$$\begin{aligned} C_i^o - C_i + \tau * r_i &= 0 \\ C_i^o &= \frac{I_i}{V} \\ C_i &= \frac{E_{\mathrm{vap},i}}{V} \end{aligned} \tag{A.18}$$

These initial conditions can be used to solve for all the reactor outlet concentrations for a given τ. These initial conditions can be used to solve the system of ODEs for the flow rates of each component. With the concentration, selectivity and conversion can be calculated as follows Integration over a range of volumes produces the flow rates for each component across the reactor, which can then be used to calculate selectivity and conversion as follows

$$X = \frac{C_O^o - C_O}{C_i^o} \qquad S = \frac{C_{\mathrm{DMC}}}{C_O^o - C_O} \tag{A.19}$$

The flows into the reactor can be calculated using the selectivity and conversion as follows

$$\begin{aligned} I_M &= \frac{\mathrm{MR}_M P_{\mathrm{D}}}{2S\mathrm{X}} \\ I_{\mathrm{CM}} &= \frac{\mathrm{MR}_{\mathrm{CM}} P_{\mathrm{D}}}{2S\mathrm{X}} \\ I_O &= \frac{P_{\mathrm{D}}}{2S\mathrm{X}} \end{aligned} \tag{A.20}$$

The volumetric flow rate, q, can be calculated as follows

$$q = \Sigma\left(\frac{F_i}{\rho_i}\right) \tag{A.21}$$

The volume of the reactor is obtained as follows

$$V = \tau * q \tag{A.22}$$

A.4: Product Component Destinations

Table A.4 : Component Classifications and Destinations

Component	Classification	Destination
Oxygen	Primary Reactant	Recycle Stream back to Mixer
Carbon Monoxide	Primary Reactant	Recycle Stream back to Mixer
Methanol	Primary Reactant	Recycle Stream back to Mixer
Dimethyl Carbonate	Primary Product	Sold as Dimethyl Carbonate
Water	Co-Product	Waste Water Stream
Carbon Dioxide	By-Product	Scrubber Exit Stream

A.5: Consistency Checks

A.5.1: Reactor Feed

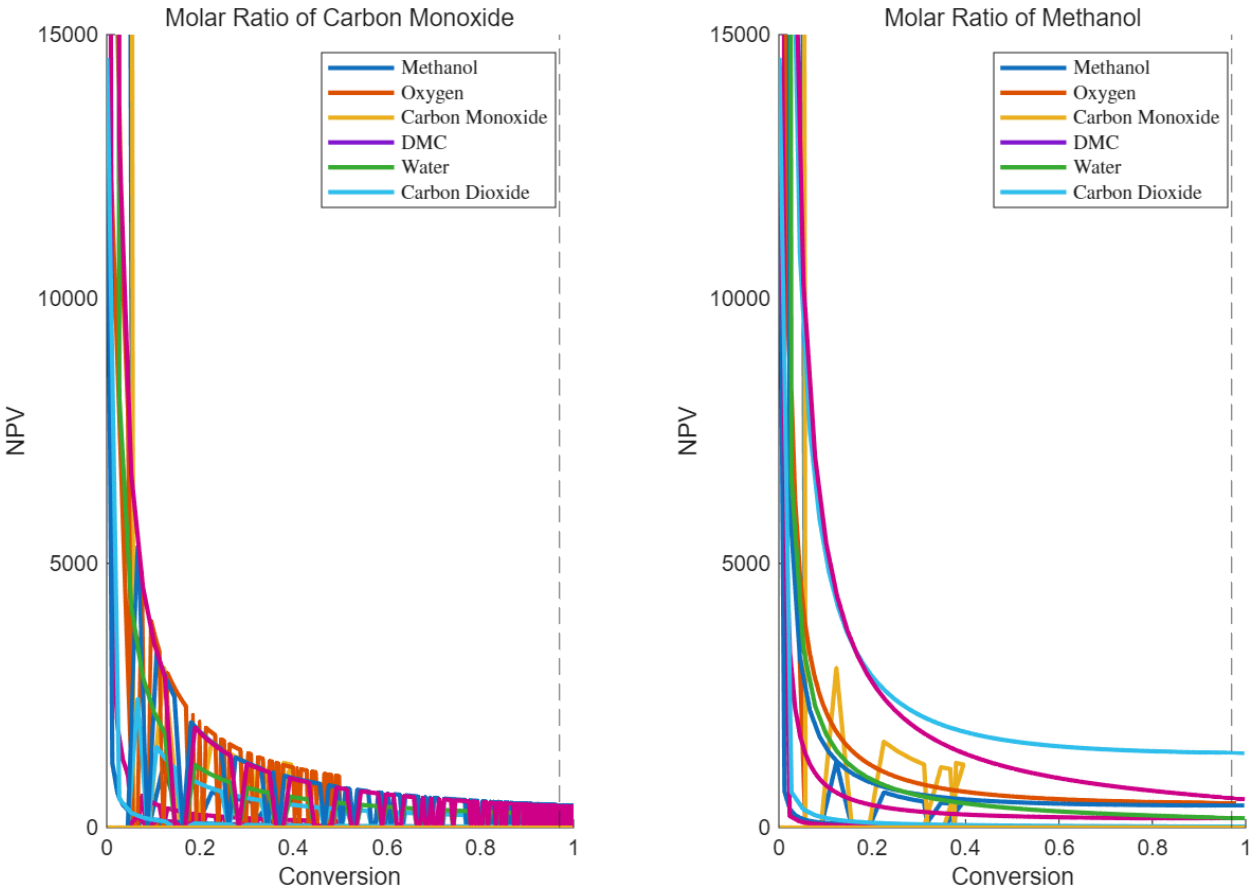


Figure A.10 : otal flow rate into the reactor versus reactor conversion at operating conditions.

A.5.2: Reactor Effluent

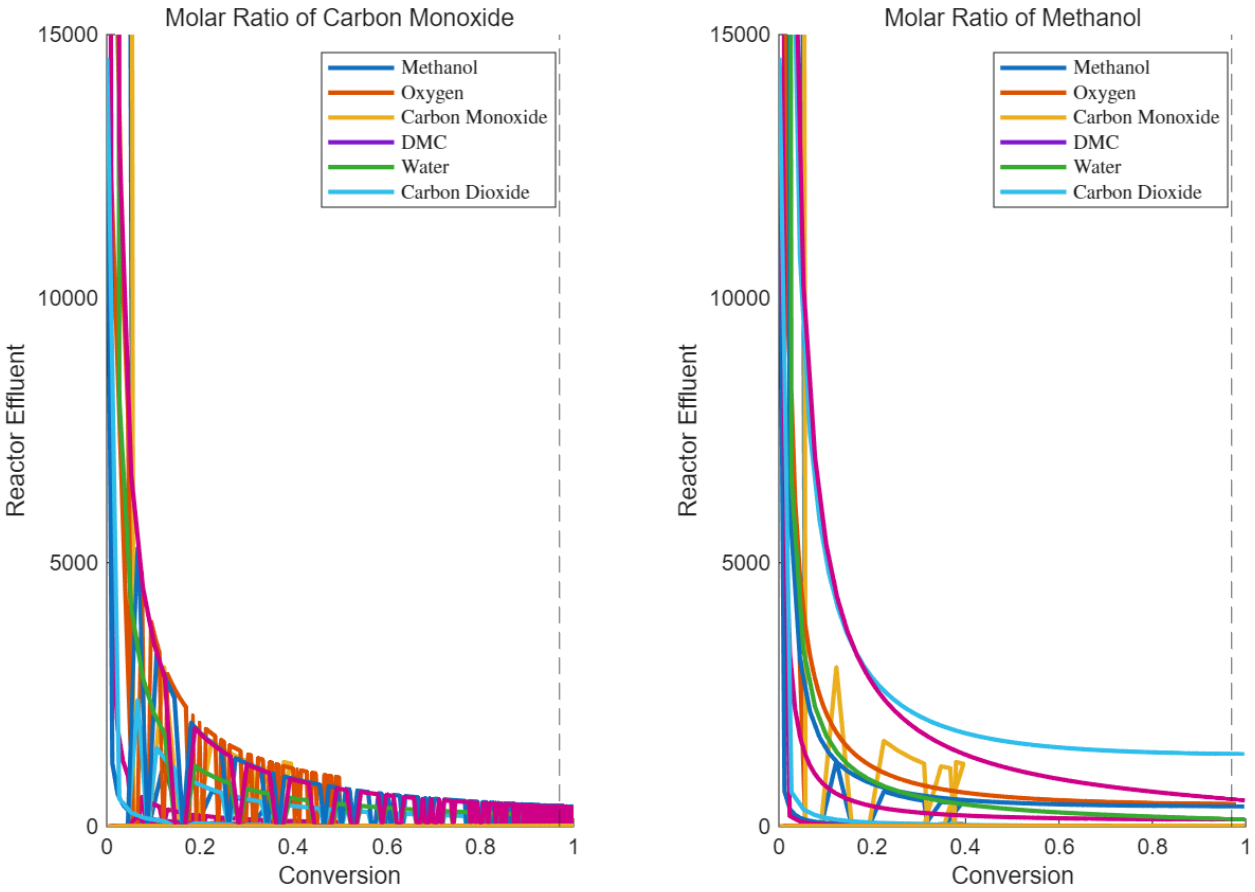


Figure A.11 : Total flow rate out of the reactor versus reactor conversion at operating condition.

A.5.3: Recycle Flowrates

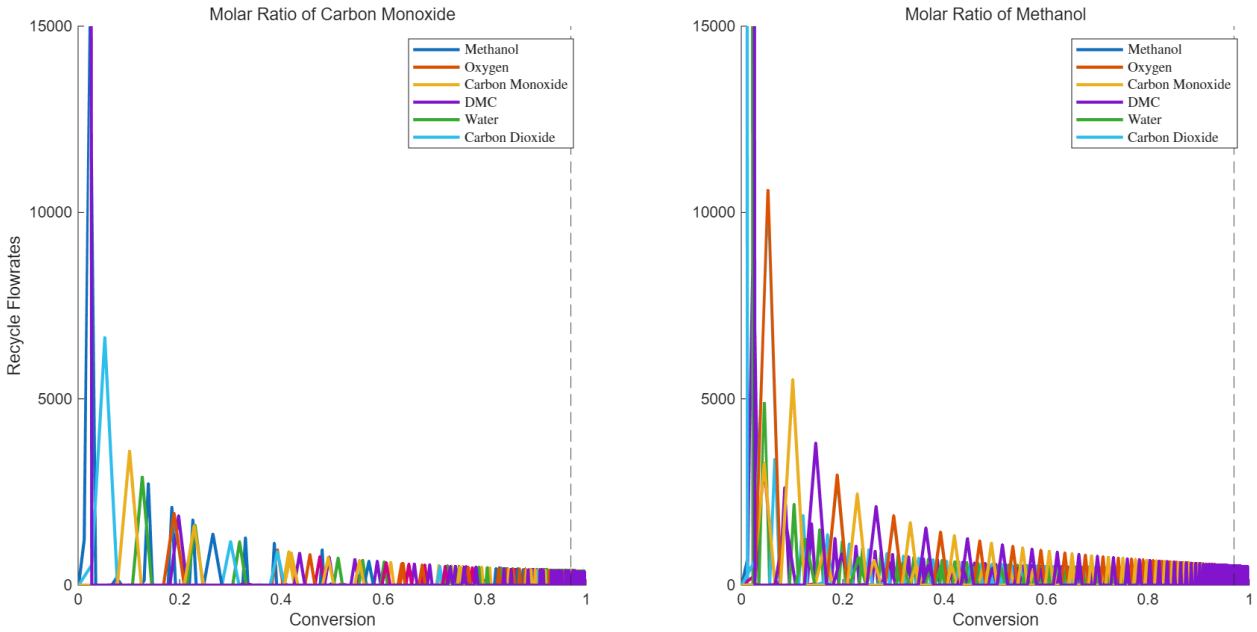


Figure A.12 : Flowrates of components out of the reactor

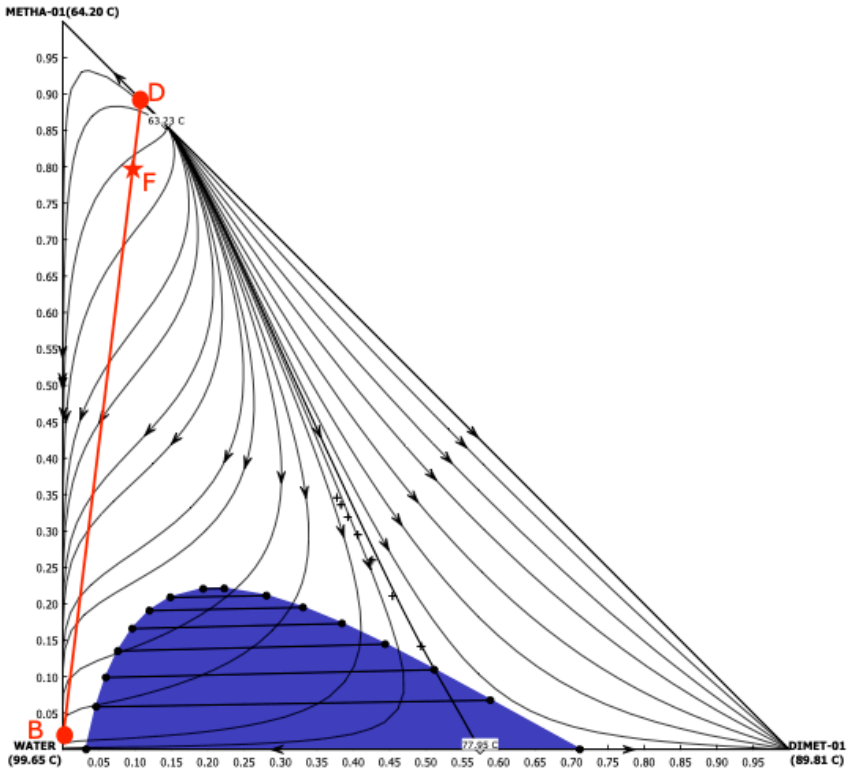
Appendix B: Separation Design Decisions

The ternary system of dimethyl carbonate, water and methanol remaining post-flash consists of 2 azeotropes and a liquid-liquid region. Distillation parameters at 1 bar for this mixture are shown in the tables below:

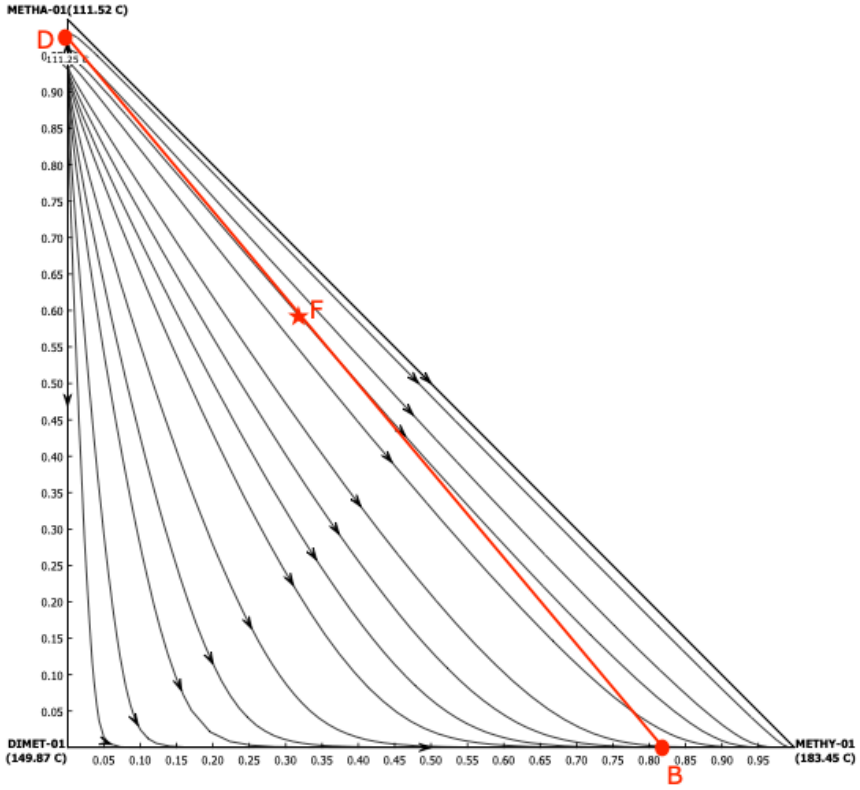
Table B.5 : Distillation Parameters at 1 Bar for a DMC-MeOH-H2O Mixture

COMP ID	Component	Classification	Temperature (°C)
DIMET-01	Dimethyl Carbonate $[C_3H_6O_3]$	Stable Node	90.22
WATER	Water $[H_2O]$	Stable Node	100.02
METHA-01	Methanol $[CH_3OH]$	Saddle	64.53

There are two azeotropes present in this mixture: a homogeneous azeotrope between dimethyl carbonate and water, and a heterogeneous azeotrope between dimethyl carbonate and methanol. As the homogeneous azeotrope is the more simple azeotrope to break, our separation system first separates water from dimethyl carbonate and methanol. Based on the inlet feed to the first distillation column, distillate and bottoms compositions were determined and then optimized to produce high purity water in the bottoms, ensuring 99.99% recovery of dimethyl carbonate to the distillate. The ternary diagram for this system is:



Based on the ConSep software on Aspen Plus, these splits were produced using a distillation column with 42 equilibrium stages and a feed stage at tray 41. These parameters were reevaluated in ASPEN HYSYS, resulting in a reflux ratio of 1.0 and reboil ratio of 15.40. Once water was removed from the system, an entraining liquid was then selected to break the methanol-DMC azeotrope. Methyl isobutyl ketone (MIBK) was selected as the entrainer due to favorable usage metrics and effective separation abilities as outlined in Matsuda et al. (2011)[9]. Using an optimized inlet ratio of DMC, methanol, and MIBK at 5 bar, the following ternary diagram is produced, where high-purity methanol is produced in the distillate, recovering 97% of the inlet DMC. The methanol produced from this column is recycled back to the reactor and combined with fresh methanol feed. A pressure above atmospheric is used in this column because it produces more favorable separation.



The DMC-MeOH-MIBK distillation column consists of 60 equilibrium stages and MIBK and MeOH-DMC feed stages at trays 33 and 47. The column has a reflux ratio of 1 and a reboil ratio of 4.19 in ASPEN HYSYS. Once the methanol was removed from the system, the DMC was able to be purified to 99.8 wt%. The column for this separation was operated at 7 bar for improved separation. The DMC-MIBK distillation column has 50 equilibrium stages, a feed stage at tray 25, a reflux ratio of 6 and a reboil ratio of 1.63. In this separation system, the MIBK is recycled back to the 2nd distillation column, where leakages and trace amounts in the DMC outlet are accounted for.

Appendix C: Heat Exchanger Network Decisions

To reduce operating costs as well as carbon dioxide output, a heat integration network was developed. This network was designed based on a stream table of of available hot and cold streams for heat exchanging. Once these heat exchangers were determined, additional heat integration was performed between distillation columns for further energy savings. The heat exchangers where 2 process streams make contact are outlined in Table C.6.

Table C.6 : Energy Savings from Process Heat Exchangers

Cool Stream (#)	Hot Stream (#)	Energy Saved (MW)
Combined Gas (5)	Decompressed Liquid (10)	0.85
3rd Column Inlet (28)	MIBK Outlet (32)	0.47

To reduce energy consumption within the separation system, a distillation column heat integration network was developed. By coupling column reboilers with condensers and using their working streams to heat and cool each other, energy usage can be optimized. This network was developed with use of section 7.6 of Doherty & Malone's *Conceptual Design of Distillation Systems*. To determine whether columns could have their units coupled, heat exchangers were created using the smaller heat duty between the 2 units. Trim heaters and coolers were then added to ensure sufficient heating and cooling of working fluid loops. To confirm that inlet and outlet temperatures did not exceed a log mean temperature difference (LMTD) of 120°C, model heat exchangers for the working fluid loops were created via ASPEN HYSYS. Both separation system heat exchangers were confirmed viable. The distillation schematic is shown in Figure , where the heat exchange between working fluids is shown. For the heat exchanger between Condenser 1 and Reboiler 2, a 1.44 MW trim heater is implemented to account for the difference in heat duty between the units and heat the reboiler working fluid to its desired temperature. For the heat exchanger between Condenser 2 and Reboiler 3, a 5.72 MW trim cooler is inserted to fully cool the working fluid before it returns to the condenser. This heat integration network allows us to greatly reduce energy usage within the separation system.

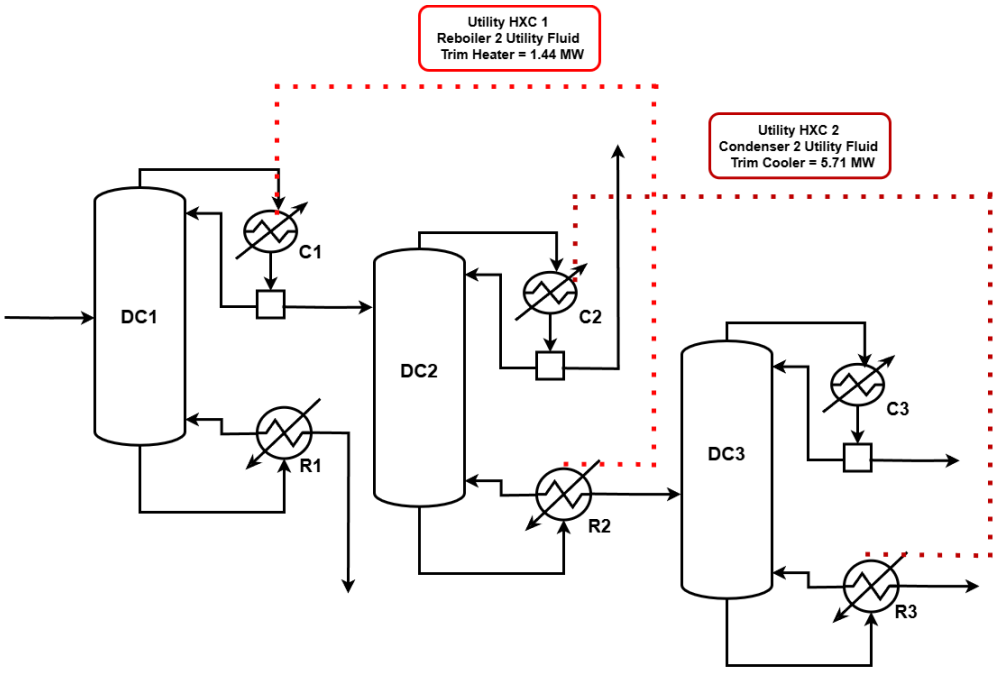


Figure C.15 : Heat Integration Network for the DMC process distillation column system.

As the reboiler attached to the first column cannot be integrated into the separation system heat network, we propose the implementation of a heat integration network between Reboiler 1 and the CSTR. The process reactor is jacketed, meaning a stream of cooling water flows through the jacket in order to keep the reactor isothermal and operating at 80 °C. Thus, the cooling working fluid removes an immense amount of heat energy from the reactor, which can then be repurposed in heating the reboiler working fluid. The energy removed from the reactor is 10.78 MW, and the duty of Reboiler 1 is 11.57 MW, meaning a 0.79 trim heater needed to be added to this heat exchange network. Heat exchanger designs are shown in Figure , where individual exchangers have been prepared for each set of exchanging units.

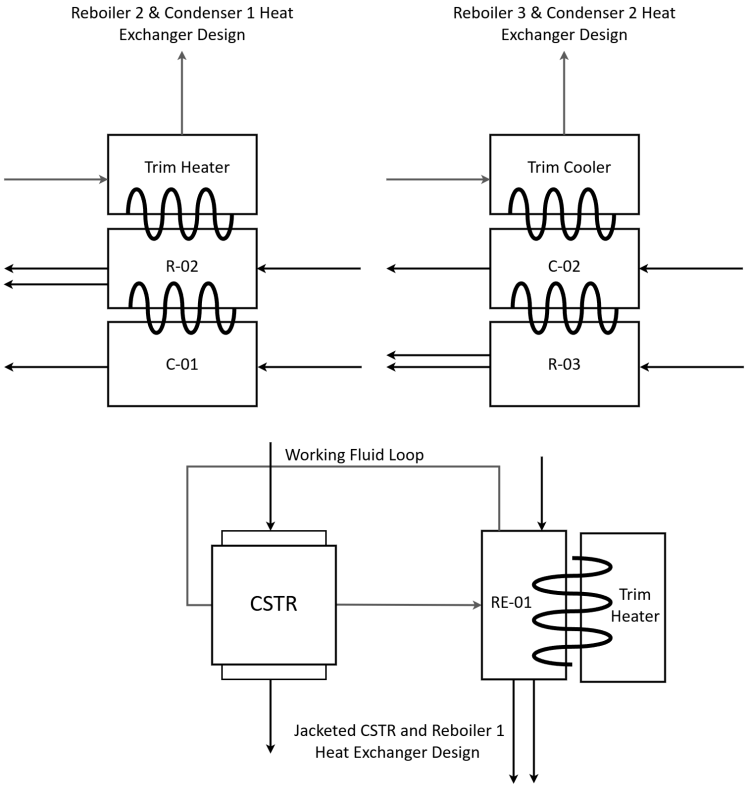


Figure C.16 : Heat Exchanger Design for plant heat exchanger network. Heat exchangers where 2 process streams make contact will implement a shell-and-tube heat exchanger.

To cool exiting and recycle streams around the plant, trim coolers with cooling water were used. These coolers operate with a cooling water inlet temperature of 35°C and a cooling water outlet temperature of 45°C.

Appendix D: Economic Analysis

D.1: Assumptions and Constants

- There is a 3 year construction period for the plant.
- Assets will depreciate linearly over 10 years.
- The plant will have a life cycle of 15 years.
- All annual income and expenses are accounted at the end of the year.
- The total tax rate on profits is 27%.

- Administration and General Services (AGS) costs approximately 5% of annual revenue.
- The Chemical Engineering Plant Cost Index in 2025 for cost correlations is 800.
- The enterprise rate is 15% annually.
- The charge for carbon emissions is accounted as a cost and is not taxed.
- Following the life cycle of the plant, the salvage value of equipment is 5%.

D.2: Component Costs

The prices of dimethyl carbonate, methanol, oxygen, and carbon monoxide as commodity chemicals were calculated from current market prices.

Table D.7 : Process Components and their Costs

Components	Cost per Tonne
Dimethyl Carbonate $[C_3H_6O_3]$	\$900/MT
Methanol $[CH_3OH]$	\$400/MT
Carbon Monoxide $[CO]$	\$150/MT
Oxygen $[O_2]$	\$100/MT
Methyl Isobutyl Ketone $[C_6H_{12}O]$	\$2560/MT[24]

D.3: Equations for Economic Analysis

D.3.1: Inside and Outside Battery Limits

The ISBL is the total installation cost for all required equipment for the chemical components of the plant, and IC is the installation cost for each separate unit of the plant.

$$\text{ISBL} = \text{I}C_{\text{heat exchangers}} + \text{I}C_{\text{reactor}} + \text{I}C_{\text{separation system}} + \text{I}C_{\text{compressor}} \tag{D.1}$$

The OSBL is the cost of installation for infrastructure outside of the chemical units of the plant, including loading facilities, administrative offices, storage tanks, power generation and waste treatment. The relation between the ISBL and the OSBL is defined as:

$$\text{OSBL} = 0.4\text{ISBL} \tag{D.2}$$

D.3.2: Direct and Indirect Costs

Direct costs refer to the sum of all expenses that can be tied to a product or specified aspect of the project, while indirect costs cover all expenses not tied to one. They are calculated using the formulas:

$$\text{Direct Costs} = \text{ISBL} + \text{OSBL} \tag{D.3}$$

$$\text{Indirect Costs} = 0.3(\text{ISBL} + \text{OSBL}) \tag{D.4}$$

D.3.3: Fixed Capital Investment

The fixed capital investment covers additional money as a possibility, which is approximately 25% for this project. The FCI can then be determined by the relation:

$$\text{FCI} = (1 + \text{Contingency}) \setminus (\text{Direct Costs} + \text{Indirect Costs}) \tag{D.5}$$

Inputting the relation involving ISBL produces

$$\text{FCI} = 2.28\text{ISBL} \tag{D.6}$$

D.3.4: Total Capital Investment

The total capital investment is calculated as

$$\text{TCI} = \text{FCI} + \text{WC} + \text{SU} \tag{D.7}$$

where working capital WC is costed as 2 months of feed, and the start-up cost SU is 10% of the FCI.

D.3.5: Revenue and Operating Cost

The revenue of the plant is defined as the profit earned from selling the streams produced. Prices used are taken from Table B.1. The revenue equation is then

$$R = \sum_{i=1}^{\text{components}} P_i \cdot c_i \tag{D.8}$$

where P_i is the product flow rate for each component and c_i is the component price. The plant operating cost is the total of the feedstock prices, the equipment operating costs within the ISBL, Administrative & General Services (AGS), and the CO_2 fee, summed as

$$\begin{aligned}\text{COM} = \text{OP}_{\text{heater}} + \text{OP}_{\text{reactor}} + \text{OP}_{\text{separation system}} + \text{OP}_{\text{compressor}} + \text{AGS} + C_{\text{C0}_2} + \\ \sum_{i=1}^{\text{components}} P_i * c_i\end{aligned} \tag{D.9}$$

D.3.6: Asset Depreciation

Using the assumption that company assets will linearly depreciate over a 10-year span, the annual depreciation can be calculated using the relation

$$d = \frac{\text{FCI - SU + Salvage}}{10} \tag{D.10}$$

where the salvage value is stated to be 5% of the fixed capital investment.

D.3.7: Cash Flows

Annual post-tax cash flows are determined by the equation

$$\text{ATCF} = (R - \text{COM} - d)(1 - t) + d \tag{D.11}$$

where t is equal to the sum of federal, state, and local tax rates and assumed to be 27%. The discounted cash flow is based on a discount rate of 15% and for each year i it is calculated as

$$\mathrm{DCF}_i = \frac{\mathrm{ATCF}}{(1+r)^i} \tag{D.12}$$

D.3.8: Net Present Value and Internal Rate of Return

The net present value, or NPV for the plant is determined from the sum of each discounted cash flow over the plant lifetime, as

$$\mathrm{NPV} = \sum_{i=0}^{\text{plant lifetime}} \mathrm{DCF}_i \tag{D.13}$$

The internal rate of return, or IRR is the discount rate r where NPV is 0 in the equation for DCF_i.

D.4: Capital and Operating Expenses Calculations

All equipment cost correlations are from Appendix E of Douglas' *Conceptual Design of Chemical Processes*.

D.4.1: Furnace Installed Cost

To determine the cost of the start-up furnace, we calculated the heat load, as this furnace must increase the temperature of the incoming stream of methanol, carbon monoxide, and oxygen to the temperature of the reactor T_R while maintaining an isothermal reactor. The total heat duty is

$$\begin{aligned} Q = F_{\mathrm{CH_4O}}\left(\Delta H_{\mathrm{vap,CH_4O}} + C_{p,\mathrm{CH_4O}}\left(T_R - T_{\mathrm{in,CH_4O}}\right)\right) + \\ F_{\mathrm{CO}}\left(C_{p,\mathrm{CO}}\right) - \left(T_R - T_{\mathrm{in,CO}}\right) \end{aligned} \tag{D.14}$$

where the heat of vaporization of methanol ΔH_{vap} = 37600 J/mol, the C_p of methanol and the combined feed of carbon monoxide and oxygen gas are 79.5 J/mol*K and 29.22 J/mol*K respectively[25–27]. The heat duty is then converted to MM Btu / hour to calculate the installed cost of the heater $\mathrm{IC}_{\mathrm{Heater}}$ using the equation

$$\mathrm{IC}_{\mathrm{Heater}} = \left(\frac{\mathrm{CEPCI}}{113.7}\right)(5.52{*}10^3)Q^{0.85(1.27+F_c)} \tag{D.15}$$

where F_c is a correction factor calculated from the design type, F_d the radiant tube material, F_m, and the design pressure, F_p.

F_d = Design Type: A-Frame Furnace

F_m = Radiant Tube Material: 321 Stainless Steel

F_p = Design Pressure

$F_c = F_d + F_m + F_p$

D.4.2: Reactor Installed Cost

The cost of the reactor, modeled as a pressure vessel was determined using the vessel dimensions with the assumption that it is a cylinder with a height 5 times greater than its diameter. The cost correlation for the installed cost of the reactor, $\mathrm{IC}_{\mathrm{reactor}}$ is

$$\mathrm{IC}_{\mathrm{reactor}} = \left(\frac{\mathrm{CEPCI}}{113.7}\right)101.9D^{1.066}H^{0.82(2.18+F_c)} \tag{D.16}$$

Where F_c, the correction factor is

F_m = Shell Material: Stainless Steel

F_p = Design Pressure F_c = F_m *F_p

D.5: Distillation Tower Installed Cost

The overall installed cost for the distillation column includes the installed cost of the tower, modeled as a pressure vessel, the trays within the tower, and the condenser and reboiler. Although the pressure vessel cost is calculated using the reactor installed cost correlation, the height and diameter are defined by the number of stages and vapor flow rate through the tower rather than a fixed ratio.

Using the condenser and reboiler heat duties and the estimated heat transfer coefficient from Table 6.2 in Doherty & Malone , the total and areas of heat exchangers needed for each portion of the distillation column were calculated. The installed cost of all the heat exchangers was then used to determine the installed cost for the condenser and reboiler. The installed cost of the distillation tower trays is calculated from

$$\mathrm{IC}_{\mathrm{trays}} = \left(\frac{\mathrm{CEPCI}}{113.7}\right) 4.7 D^{1.55} H F_c \tag{D.17}$$

Where F_c is a correction factor calculated from F_s, determined by the tray spacing F_m, determined by the tray material, and F_t, determined by the tray type.

D.5.1: Reactor Operating Cost

The operating cost of the reactor is determined using the cost of cooling utilities to maintain isothermal operation of the reactor. Using the $2/GJ value for cooling water, the total operating cost for the reactor is

$$\mathrm{OP}_{\mathrm{reactor}} = \dot{Q}\left[\frac{\mathrm{GJ}}{\mathrm{Year}}\right] * \text{ Cooling Water Cost} \tag{D.18}$$

D.6: Compressor Installed Cost

The compressor installed cost is dependant on the power of the compressor. The type of compressor used in this process is a centrifugal motor compressor, with a correction factor F_c of 1.0. The total installed cost for a compressor is

$$\mathrm{IC}_{\mathrm{compressor}} = \left(\frac{\mathrm{CEPCI}}{113.7}\right) (517.5)(\mathrm{W}\ [\mathrm{GJ}])^{0.82} F_c \tag{D.19}$$

D.6.1: Compressor Operating Cost

The compressor operating cost is calculated from its power given from HYSYS. Under the assumption that compressors are powered via electricity, and the price of electricity is $0.08/kWh, the compressor operating cost is

$$\mathrm{OP}_{\mathrm{compressor}} = \mathrm{W} * \text{price of electricity} \tag{D.20}$$

D.7: Heat Exchanger Operating Cost

The operating cost for a heat exchanger where 2 process streams make contact with each other is zero. The operating cost of the heat exchanger is taken into account if a process stream is contacted by an

external cold or hot stream. If the process stream makes contact with a hot stream, the operating cost of the heat exchanger is calculated by the heat duty production via natural gas combustion producing the equation

$$\mathrm{OP}_{\text{heat exchanger}} = \dot{Q} * (\text{fuel price}) \tag{D.21}$$

To determine the cooler costs, the price of the coolant is calculated as a function of the temperature of the coolant

$$\text{Cost } \left(\frac{\$}{\mathrm{GJ}}\right) = \varepsilon(0.047 * T^2 - 2.97T + 105.88) \tag{D.22}$$

where T is in °C and ε is the price of electricity in $/kWh. At temperatures between and 5 and 35°C, chilled water with a cost per GJ of $4.9 is used.

D.8: Distillation Operating Cost

The operating cost for a distillation column is the sum of the operating costs of the reboiler, acting as a heater, and the condenser, acting as a cooler. Cost correlations are as described in Appendix C.7.

D.8.1: Minimum Separation Power

The minimum separation power can be derived from the exit streams of each component entering the separation system, as defined in Table A1. Assuming temperature and pressure are constant, the minimum work, W_{min} to overcome ΔG of mixing is

$$\dot{\mathrm{W}}_{\text{min}} = \Sigma_{j=1}^{\text{exit streams}} \left(F_j \Sigma_{i=1}^{\text{components}} x_{\mathrm{i,j}} \ln\left(\frac{x_{\mathrm{i,j}}}{z_i}\right)\right) \tag{D.23}$$

D.8.2: Carbon Charge of Process

The carbon charge for this process is stated as $\$\frac{100}{\mathrm{MT}}$ and can be converted to a charge in terms of energy used

$$\frac{\$100}{\mathrm{MT}} * \frac{1\mathrm{MT}}{10^3 \text{ kg}} * \frac{44.01 \text{ kg}}{1 \text{ kgmol}} * \frac{1 \text{ kgmol } \mathrm{CO_2}}{1 \text{ kgmol } \mathrm{CH_4}} * \frac{1 \text{ kgmol } \mathrm{CH_4}}{1.125 \text{ GJ}} = \$\frac{5.49}{\mathrm{GJ}} \tag{D.24}$$

Scope 1 emissions consist of all carbon dioxide produced as a by-product of the secondary reaction. Scope 2 emissions consist of the energy used to power the compressor and trim heaters within the plant.

D.9: Entrainer Purchase Cost

In order to account for the cost of purchasing methyl isobutyl ketone (MIBK), a point purchase calculation was devised to account for the recycle loop MIBK follows during separation in the plant distillation columns. To calculate the amount of MIBK required for the recycle loop, we divided the column volume by the volumetric flow rate inlet to the column to get the residence time, τ of MIBK in the column. This was done for both DC 2 and DC 3 where MIBK enters and exits the separation system. The residence times were summed together and multiplied by the MIBK inlet flow rate in MT/hr to get the purchase cost of the entrainer. The price per MT of MIBK is listed in Table D.8.

D.10: Wastewater Treatment

The price of wastewater treatment varies based on the type of treatment necessary for wastewater. Our wastewater is not radioactive, and in our separations system, does not leave the plant meeting emissions standards of below 30 ppm of mixed organic compounds. Therefore, our wastewater requires tertiary treatment[28]. The price of wastewater treatment in \$/m^3 is

$$C_{\text{t,ww}} = a(\text{CEPCI}_{2025}) + \text{bC}_f \tag{D.25}$$

where CEPCI is a capital cost index that adjusts for inflation and C_f is the price of fuel in \$/GJ. The factors a and b are calculated by

$$a = 0.001 + 2 * 10^{-4} q^{-1} \qquad b = 0.1 \tag{D.26}$$

where q is the liquid flow rate of water leaving the plant in m^3/s. Converting our molar flow rate of water to a volumetric flow rate of liquid water, then multiplying by this cost per volume of wastewater gives our total cost of wastewater treatment for the plant. This cost is extremely small, so it is not included as a component of the operating cost.

D.11: Sizing Flash Drums

Flash drums were designed as pressure vessels with a diameter-to-height ratio of 1:4. The residence time of the vessel was set at 15 minutes, so the volume of the flash drum was calculated as

$$V = \dot{Q} * \tau \tag{D.27}$$

where V is the flash drum volume in m^3, $\dot{Q}$ is the volumetric flow rate entering the flash in $\frac{m^3}{s}$, and τ is the residence time in seconds.

D.12: Sizing Compressors

The braking horsepower (bhp) of the compressor was calculated from values of streams from HYSYS. Using equation 6.5-1 from Doherty & Malone's *Conceptual Design of Distillation Systems* ,

$$\text{bhp} = \left(\frac{1}{0.8}\right)\frac{3.03 * 10^{-}5}{\gamma} P_{\text{in}} Q_{\text{in}} \left[\left(\frac{T_{\text{out}}}{T_{\text{in}}}\right) - 1\right] \tag{D.28}$$

where P_{in} and P_{out} are pressures in $\frac{\text{lbf}}{\text{ft}^2}$, Q_{in} is in $\frac{\text{ft}^3}{\text{min}}$, and the ratio of absolute temperatures and ratio of specific heat γ are

$$\frac{T_{\text{out}}}{T_{\text{in}}} = \left(\frac{P_{\text{out}}}{P_{\text{in}}}\right)^{\gamma} \qquad \gamma = \frac{R}{C_p} \tag{D.29}$$

where C_p is the heat capacity of the stream at constant pressure.

D.13: Sizing Heat Exchangers

From the necessary heat duties and the estimated heat transfer coefficients in Table 6.2 of Doherty & Malone , we can calculate the areas of the heat exchangers.

The heat transfer coefficients for coolers was determined via the required coolant temperature. We assumed the cold side of heat exchanger would be liquid water or high pressure steam.

The heat transfer coefficients for heaters was determined via the required hot stream temperature. Condensing steam was used for hot streams in these heat exchangers.

To remain within the sizing range of the correlations in Appendix D for the installed and operating costs of heaters and coolers within this system, the maximum heat exchanger size was set to be 2,300 m^3. If the heat exchanger was found to be larger than this, it was divided into a series of heat exchangers. Heat exchangers did not exceed a temperature difference of 120°C to ensure safe operation and remain within material constraints, unless the material provided (Grade 304/316 SS) was proven to adequately withstand extreme thermal environments.

D.14: Estimating Scrubber & Acid Removal System Costs

To evaluate potential installation costs for the scrubber and stabilizer in our MATLAB simulation, the minimum work $\dot{W}_{\text{min}}$ was multiplied by the estimated ratio of the real work to the minimum work, γ and the capital cost correlation factor, $C_{\$/\text{W}}$ allowing us to account for the necessary costs of both process units.

$$\text{IC}_{\text{Sc/St}} = \dot{W}_{\text{min}} * \gamma * C_{\$/\text{W}} \tag{D.30}$$

D.15: Cash Flow Analysis

D.15.1: Discounted & Non-discounted Cash Flow Graphs

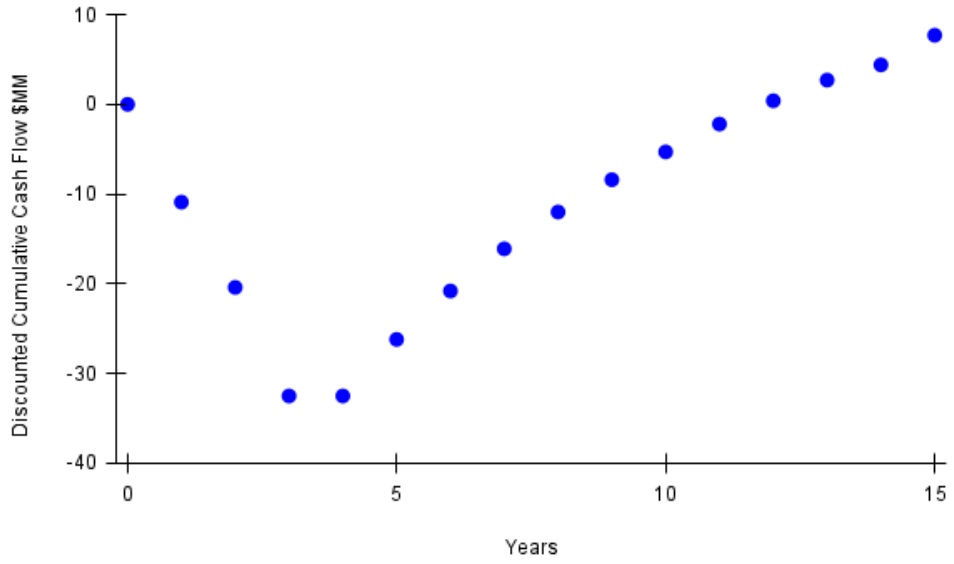


Figure B.1: The cumulative discounted cash flow for the plant over its construction and lifetime.

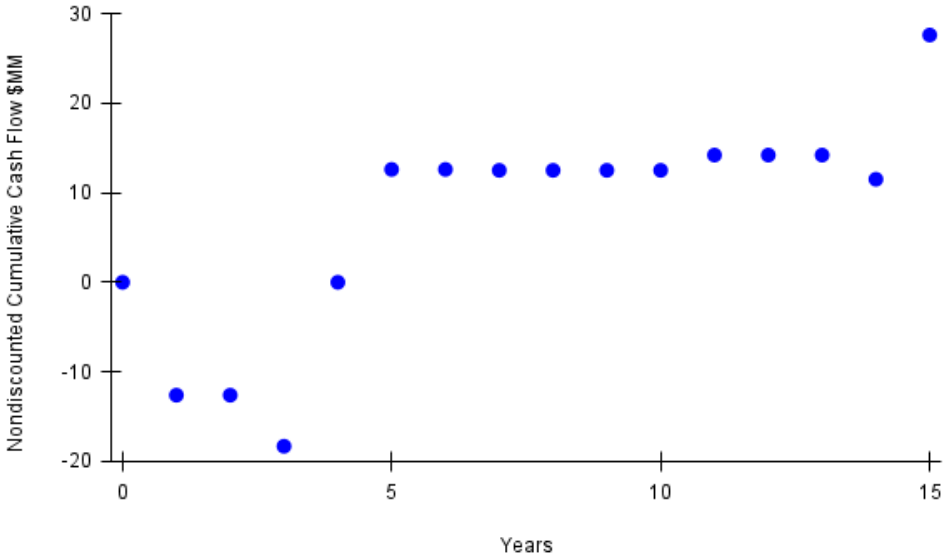


Figure B.2: The cumulative non-discounted cash flow for the plant over its construction and lifetime.

D.15.2: Discounted Cash Flow Spreadsheet

BICC, Inc.
123 Lagoon Road
Santa Barbara, CA

Project Name: FEL-1 DMC Plant

Rev	Date	BY	APVD	Rev	Date	BY	APVD
V1.12	4/30/25	D9					

ECONOMIC ANALYSIS

Plant Location: Mount Vernon, IN

Case Description: DMC Production via Oxidative Carbonylation

REVENUES AND PRODUCTION COSTS

	$MM/yr
Main product revenue	44.6
Byproduct revenue	0.0
Raw materials cost	21.8
Utilities cost	1.3
Consumables cost	0.0
CO_2 sustainability charge	3.5
VCOP	26.6
Salary and overheads	0.0
Maintenance	0.0
Interest	0.0
AGS	2.2
FCOP	2.2

CAPITAL COSTS

	$MM
ISBL Capital Cost	16.7
OSBL Capital Cost	6.7
Indirect Cost	7.0
Contingency	4.2
Total Fixed Capital Cost	38.1
Working Capital	3.6
Start-up Costs	2.2
Land	0.0
Total Capital Investment	43.8

CONSTRUCTION SCHEDULE

Year	% FC	% WC	% SU	% FCOP	% VCOP
0					
1	33%				
2	33%				
3	33%	100%	100%		
4				100%	100%
5				100%	100%
6+				100%	100%

ECONOMIC ASSUMPTIONS

On Stream	8400 hr/yr	Discount Rate	15%	Depreciation method	Straight-Line
	350 day/yr	Tax Rate	27%	Depreciation period	10 yrs
Project Life	15 yr	Salvage Value	5%		

CASH FLOW ANALYSIS

All figures in $MM unless indicated

Project Year	Cap. Ex.	Revenue	COM	Gr. Profit	Deprcn.	Taxable Inc	Taxes Paid	Cash Flow	PV of CF	NPV
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	12.6	0.0	0.0	0.0	0.0	0.0	0.0	-12.6	-10.9	-10.9
2	12.6	0.0	0.0	0.0	0.0	0.0	0.0	-12.6	-9.5	-20.4
3	18.3	0.0	0.0	0.0	0.0	0.0	0.0	-18.3	-12.0	-32.5
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-32.5
5	0.0	44.6	28.8	15.8	3.8	12.0	3.2	12.6	6.2	-26.2
6	0.0	44.6	28.8	15.8	3.8	12.0	3.2	12.6	5.4	-20.8
7	0.0	44.6	28.8	15.8	3.8	11.9	3.2	12.5	4.7	-16.1
8	0.0	44.6	28.8	15.8	3.8	11.9	3.2	12.5	4.1	-12.0
9	0.0	44.6	28.8	15.8	3.8	11.9	3.2	12.5	3.6	-8.4
10	0.0	44.6	28.8	15.8	3.8	11.9	3.2	12.5	3.1	-5.3
11	0.0	44.6	28.8	15.8	10.1	5.7	1.5	14.2	3.1	-2.2
12	0.0	44.6	28.8	15.8	10.1	5.7	1.5	14.2	2.7	0.4
13	0.0	44.6	28.8	15.8	10.1	5.7	1.5	14.2	2.3	2.7
14	0.0	44.6	28.8	15.8	0.0	15.8	4.3	11.5	1.6	4.4
15	-4.8	49.4	28.8	20.6	0.0	20.6	5.6	27.6	3.4	7.7

ECONOMIC ANALYSIS

NPV	15 years	7.7 $MM		IRR	19%
NPV at yr	12	0.4 $MM			

NOTES

1. All cash flows are assumed to occur at the end of the project year.
2.
3.

Appendix E: Conceptual MATLAB Design vs HYSYS Analysis

E.1: MATLAB vs HYSYS Comparison and Analysis

MATLAB and HYSYS mass and energy flows for this system differ due to thermodynamic assumptions made in the MATLAB simulation. MATLAB treats all gasses as ideal and assumes that the heats of reaction are constant, while HYSYS uses the UNIQUAC model to determine complex relationships for these variables. Due to these discrepancies, the HYSYS simulation results should produce more accurate calculations.

Variable	HYSYS	MATLAB
Reactor Inlet Temperature (°C)	80	80
Reactor Outlet Temperature (°C)	80	80
Pressure (bar)	27	27
CO-to-O2 Molar Ratio	24	24
MeOH-to-O2 Molar Ratio	9	9
Reactor Volume (m^3)	37	37
Conversion	97	97
Selectivity of Dimethyl Carbonate	45	51

Table E.9 : Energy streams as modeled by our plant design

Stream	Unit Name	Energy Flow (MW)
E1	HXC 1	0.85
E2	C1-R2 Trim Heater	1.44
E3	C2-R3 Trim Cooler	5.71
E4	Water Trim Cooler	0.087
E5	MIBK Trim Cooler	2.51
E6	HXC 2	0.59
E7	Methanol Trim Cooler	0.38
E8	DMC Trim Cooler	0.41
E9	Condenser 3	3.62

Table E.10 : Equipment units as modeled by HYSYS & Heat Integration network. The mixers, pumps and expanders are not included as they are beyond the scope of this report.

Name	Unit Operation	Material	Design Parameters	Capital Cost ($MM)
CSTR	Continuous Stirred Jacketed Reactor	Teflon Lined Carbon Steel	Operates isothermally at 80°C and 27 bar, sized at 37 m^3	0.51
V-100	Liquid Flash Drum	Grade 316 Stainless Steel	Flash separates CO/O2/CO2 from DMC/MeOH/ H2O at 117.5°C. Has a volume of 6.2 m^3	0.17
V-101	Vapor Flash Drum	Grade 316 Stainless Steel	Flash separates CO/O2/CO2 from DMC/MeOH/ H2O at 50.25°C. Has a volume of 24.4 m^3	0.30
HXC 1	Heat Exchanger	Grade 316 Stainless Steel	Cools the reactor inlet and heats the reactor outlet via countercurrent exchange. Minimizes utility	0.67

Name	Unit Operation	Material	Design Parameters	Capital Cost ($MM)
			energy needed for the furnace. The heat transfer coefficient is 40.8 $\frac{W}{m^2 \cdot K}$ and surface area is 1547 m^2	
K-101	Compressor	Grade 304 Stainless Steel	Centrifugal motor compressor. Compresses recycled gas products to 27 bar before entering CSTR. 1.332 MW Power	1.9
DC 1	Distillation Column	Grade 316 Stainless Steel	Separates water from methanol and DMC using 42 stages of sieve trays with spacing of 24 in. The reflux ratio is 1.0 and reboil ratio is 14.4. Operates at 1 bar. Has an area of 94.5 m^2 and a height of 27 m	3.4
DC 2	Distillation Column	Grade 316 Stainless Steel	Separates methanol from MIBK and DMC using 55 stages of sieve trays with spacing of 18 in. The reflux ratio is 1.0 and reboil ratio is 4.19. Operates at 5 bar. Has an area of 124.5 m^2 and a height of 83 m.	5.1
DC 3	Distillation Column	Grade 316 Stainless Steel	Separates methanol from MIBK and DMC using 50 stages of sieve trays with spacing of 18 in. The reflux ratio is 6.0 and reboil ratio is 1.638. Operates at 7 bar. Has an area of 114 m^2 and a height of 76 m	2.1
HXC 2	Heat Exchanger	Grade 304 Stainless Steel	Cools the DMC outlet via countercurrent exchange with the DC 3 bottoms stream. The heat transfer coefficient is 724 $\frac{W}{m^2 \cdot K}$ and surface area is 105 m^2	0.21
Water Recovery Cooler	Heat Exchanger	Grade 316 Stainless Steel	Cools the water production outlet via countercurrent exchange with cooling water. The heat transfer coefficient is 2257.7 $\frac{W}{m^2 \cdot K}$ and surface area is 2.7 m^2	0.07

Name	Unit Operation	Material	Design Parameters	Capital Cost ($MM)
DMC Production Cooler	Heat Exchanger	Grade 304 Stainless Steel	Cools the DMC production outlet via countercurrent exchange with cooling water. The heat transfer coefficient is 1361.5 $\frac{W}{m^2 \cdot K}$ and surface area is 14.9 m^2	0.20
MIBK Recycle Cooler	Heat Exchanger	Grade 304 Stainless Steel	Cools the MIBK recycle stream via countercurrent exchange with cooling water. The heat transfer coefficient is 1042.6 $\frac{W}{m^2 \cdot K}$ and surface area is 40.0 m^2	0.16
Methanol Recycle Cooler	Heat Exchanger	Grade 304 Stainless Steel	Cools the methanol recycle stream via countercurrent exchange with cooling water. The heat transfer coefficient is 1374.6 $\frac{W}{m^2 \cdot K}$ and surface area is 4.5 m^2	0.098
Reboiler 2 Trim Heater	Heat Exchanger	Grade 304 Stainless Steel	Heats the reboiler 2 working fluid stream via countercurrent exchange with condensate steam. The heat transfer coefficient is 475.0 $\frac{W}{m^2 \cdot K}$ and surface area is 23.9 m^2	0.084
Condenser 2 Trim Cooler	Heat Exchanger	Grade 304 Stainless Steel	Cools the condenser 2 working fluid stream via countercurrent exchange with cooling water. The heat transfer coefficient is 173.3 $\frac{W}{m^2 \cdot K}$ and surface area is 924 m^2	0.085
Amine Scrubber	Scrubber	Grade 304 Stainless Steel	Scrubs carbon dioxide from gas recycle stream using amine compounds (MEA, DEA, MDEA) and produces “sweetened” gas low in CO2.	1.6

Table E.11 : HYSYS Distillation Columns: Number of Stages, Reflux & Reboiler Ratios, Feed Stage

Parameter	DC 1	DC 2	DC 3
Number of Stages	42	55	50
Reflux Ratio	1.0	1.0	6.0
Reboil Ratio	15.4	4.19	1.63
Feed Stage(s)	41	33,47	25

E.2: Heating Composite Curve

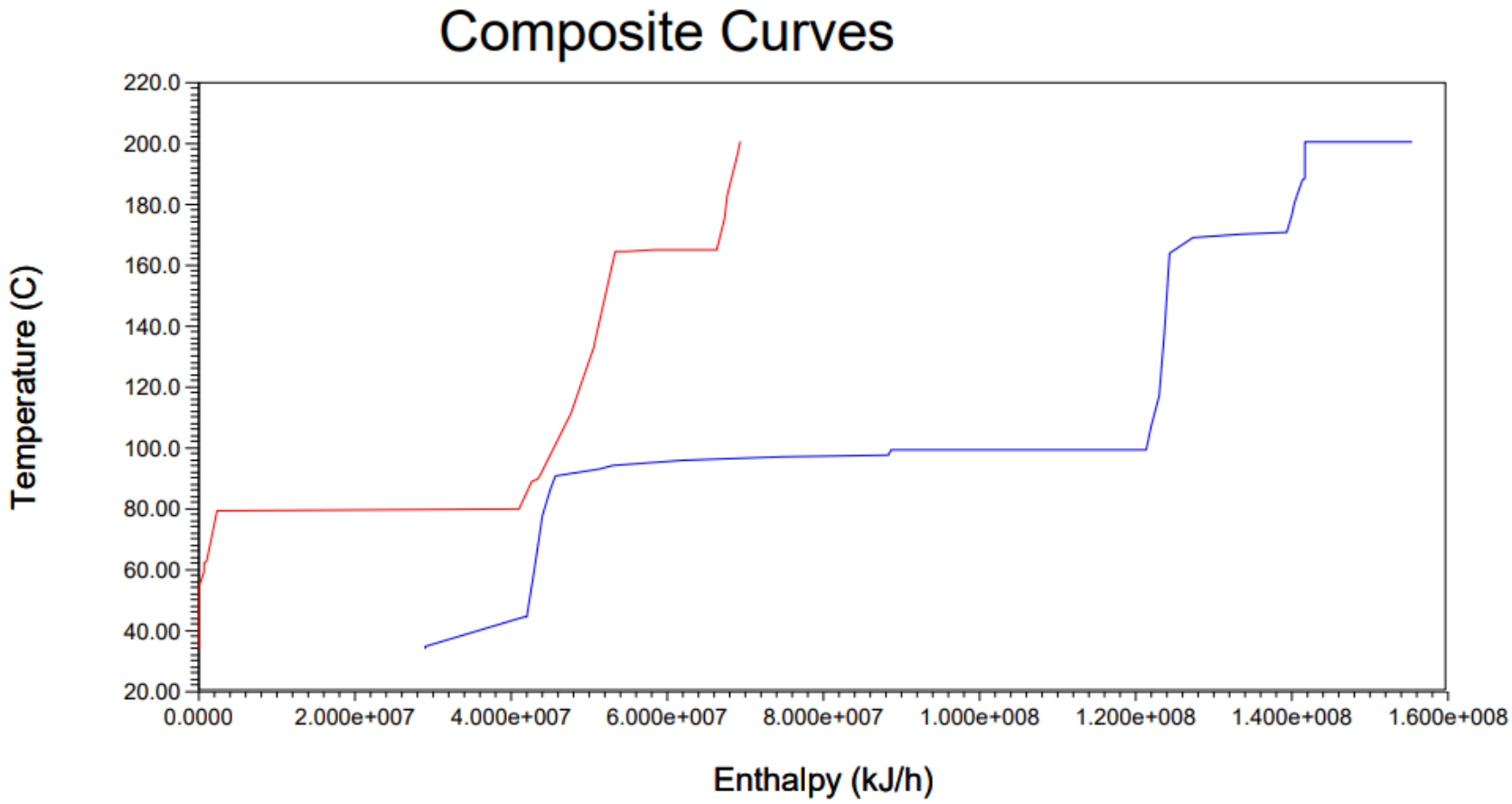


Figure E.19 : Temperature (C) vs. Enthalpy (kJ/h) Graph for Heat Exchanger Network Mapped In ASPEN HYSYS

Appendix F: HAZOP Analysis

Table E.1: Summary of safety concerns and recommendations the address these concerns for specific units within our process

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
Furnace	Fuel Flow	More	Valve fully open Incorrectly calibrated flow meter and control system for incoming flow	The furnace may overheat and over pressurize, causing an explosion	Temperature control system with a fail-close fuel feed valve, and an emergency venting and cooling system
	Fuel Flow	Less	Valve closed Blockage in pipes feeding fuel to furnace Leakage in the inlet flow	The furnace may not heat the reactants enough to drive the reaction May impact worker's health	Temperature control system with alarm and control system for reactant valve to control the fuel flow
	Air Flow	More	Issue with air intake location Incorrect air-to-fuel ratio settings	Excess air may increase availability of oxygen for combustion leading to more heat generated Increase the availability of oxygen for explosions	Air-to-fuel ratio monitoring Air feed concentration sensor that controls fuel feed valve
	Air Flow	Less	Blocked air intake	May lead to incomplete combustion, carbon monoxide formation, and risk of explosion due to unburned fuel accumulation	Air-to-fuel ratio monitoring Air feed concentration sensor that controls fuel feed valve
	Reactant Flow	More	Flow control valve failure Incorrectly calibrated flow meter	Reduced residence time in furnace leading to incomplete heating of reactants	Flow monitoring system with an automated control

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
				May stop reaction from occurring	valve and high flow alarms
	Reactant Flow	Less	Valve close Leakage in inlet stream Pipeline blockage	Reactants remain in the furnace longer, leading to potential overheating and coking of reactants With little to no reactants the furnace may overheat and, which can lead to an explosion	Control system for reactant feed valve, emergency shutdown for overheating, pressure monitoring to detect blockages
	Temperature	More	Reduced reactant feed Increase air-to-fuel ratio Increased flow rate of natural gas Blockage of inlet or exit stream of reactant	Increase the likelihood of explosion Coking formation may occur on the walls of the furnace	Temperature control system with alarms Emergency cooling system Fail-close valve on feed or gas
	Temperature	Less	Increased reactant feed Decreased air-to-fuel ratio Decreased flow rate of natural gas	Reactants do not reach the necessary temperature for reaction to occur, leading to no product formation	Temperature control system with alarm Redirect reactant feed stream to a storage Schedule maintenance for cleaning any coke formation
	Pressure	More	Increased reactant feed Increased air-to-fuel ratio Increased flow rate of natural gas	Increase the likelihood of an explosion or structural damage	Emergency pressure relief system with fail-open valve on piped connected to furnace Pressure sensors with

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
			Blockage of inlet or exit stream of reactant		automatic trip mechanisms
	Pressure	Less	Decreased reactant feed Decreased air-to-fuel ratio Decreased flow rate of natural gas	Reduces the temperature and can lead to inadequate conditions for reaction to occur	Pressure monitoring with pressure control valves
Mix-100	CO Concentration	More	Fresh CO valve fully open Blocked O_2 pipe High amounts of CO coming from recycle stream	Explosive CO-O_2 mixture	CO concentration alarm Fail-close CO feed valve on high O_2 with a fail close valve for the O_2
	CO Concentration	Less	CO feed valve closed Blocked CO pipeline Pipeline leak	Risk of carbon monoxide poisoning if the cause is a leak Higher concentrations of O_2 in mixture increasing the risk of an explosion	CO concentration alarm Backup CO pipeline with fail open valve Backup fail open inert pipeline to ensure small oxygen concentration in mixture
	O_2 Concentration	More	Oxygen fresh feed valve fully open CO fresh feed valve closed or blocked	Higher concentration of O_2 in mixture resulting in an explosive mixture Oxygen exceeds its lower explosion limit	O_2 concentration level alarms Fail-close O_2 feed valve

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	O_2 Fresh Feed Flow	Less	Fresh O_2 feed valve fully closed O_2 flow pipeline blocked	Low reaction conversion Insufficient amount of product product produced CO build up	O low level concentration alarm O_2 backup feed with fail-open valve
	Temperature	More	Recycle feed coming in excessively hot	Increase in pressure Can lead to equipment rupture	Temperature sensors with alarms Backup cooling system
	Temperature	Less	Cooler fresh feed of CO and O_2	Pressure decreases Incorrect reactor inlet pressure	Temperature sensors with alarms
	Pressure	More	Outlet blockage Excess recycle Higher recycle temperature	Over pressurized mixer increases the risk of equipment rupture Inefficient mixing Incorrect reactor inlet pressure	Pressure relief valves Pressure monitoring with alarms
	Pressure	Less	Leak in the mixer Insufficient feed pressure	Poor mixing Incorrect reactor inlet pressure	Leak detection system Pressure monitoring with alarms
	Total Flow Rate	More	Fresh feed and recycle valves fully open	Overflow the reactor Low reaction conversion	Flow meter detector with alarms
	Total Flow Rate	Less	Blocked feed pipeline Closed valves	Under-feed into the reactor Higher O_2 concentration levels in the reactor feed	Low-flow alarm Routine mixer maintenance

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
Mix-101	Methanol Flow	More	Valve fully open Excess Recycle	Floods the reactor and catalyst	Flow controller and fail-close recycle valve
	Methanol Fresh Feed Flow	Less	Blocked pipe Valve closed	Low conversion	Low-flow alarm
	Temperature	More	Recycle coming in excessively hot	Vapor-liquid ratio incorrect Incorrect reactor inlet temperature	Temperatures sensors with alarms
	Temperature	Less	Cool feed coming into the mixer	Poor mixing Incorrect reactor inlet temperature and pressure	Temperature sensors with alarms
	Pressure	More	Outlet blockage Over pressurized pump	Increases the risk of equipment rupture	Relief valve Pressure monitoring system with alarms
	Pressure	Less	Leak in the mixer Under pressurized pump	Incorrect reactor inlet pressure	Pressure monitoring system with alarms
Catalyst Bed	HCl Concentration	More	Incomplete acid removal Over production of acid	Catalyst corrosion or poisoning Material corrosion	HCl concentration sensors with alarms Use high concentration HCl corrosion resistant material
Filtration	Pressure Drop	More	Clogged filters Catalyst buildup	Overpressure	Pressure drop alarm
	Pressure Drop	Less	Leak Broken equipment	Loss of separation efficiency	Pressure drop alarm

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
Acid Remover	HCl Concentration	More	Media exhaustion Excess amount of HCl in the reactor	Downstream equipment corrosion	HCl concentration monitor with alarms Fail-close valves so HCl does not make contact with downstream equipment
CSTR	Temperature	More	Reactants coming in excessively Reactants excessively pressurized	Incorrect reactor temperature Kinetics invalid at higher temperature Dimethyl carbonate decomposes at higher temperatures Can lead to a runaway reaction	Temperature senors with alarms Fail-open backup cooling systems
	Temperature	Less	Reactants not heated to desired temperature	Poor conversion due to off spec kinetics Not enough amount of product produced	Temperature sensors with alarms
	Pressure	More	Excess reactants Reactor outlet blockage Increase in temperature	Reactor becomes over pressurized leading to equipment rupture and explosion	Pressure relief valve with fail-open vent Pressure monitoring with alarms
	Pressure	Less	Possible leak Under-pressurized reactants	Lower conversion Possible O_2 accumulation leading to O_2 concentrations to exceed the lower explosion limit	Pressure monitoring system with alarms
Flash	Temperature	More	Overheating of inlet stream	Poor vapor-liquid separation	Temperature control system with alarms

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
			Faulty temperature controller	Can cause pressure build up inside the column	
	Temperature	Less	Inlet flow is insufficiently heated	Incomplete vaporization leading to off-spec products	Temperature control system with alarms
	Pressure	More	Blockage at outlet High feed flow rate	Pressure build up may cause the vessel to explode	Pressure relief valves
	Pressure	Less	Blockage at inlet Leak in the column Decreased flow into the column	Poor separation die to lower vapor pressure	Automatic pressure control valve Continuous pressure monitoring system
	Flow Rate	More	Valve fully open Incorrect flow meter calibration	May lead to flooding and poor separation	Fail-close valve at feed line Temperature control system with alarms
	Flow Rate	Less	Valved closed Blockage in feed line	May lead to temperature rise in column	Temperature control system with alarms Continuous feed monitoring
Distillation Column	Temperature	More	Incoming feed is excessively hot Incorrect temperature controller	May lead to decomposition of components Increased pressure due to excess vapor	Temperature control system with alarms
	Temperature	Less	Fouling in reboiler	Incorrect separation of components Bottoms may be off spec	Temperature control system with alarms Routine reboiler cleaning

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	Pressure	More	Blockage in exit streams High vapor flow rate	Pressure build up may lead to column rupture	High pressure relief valve High pressure alarm system
	Pressure	Less	Leak in column	May cause poor separation due to disruption in internal flow patterns	Pressure monitoring system with alarms
	Flow Rate	More	Feed valve fully open	Overloading of distillation trays Poor separation Weeing may occur if too much feed enters the column	Flow control system between the flash and distillation column Fail-close valve on distillation feed
	Flow Rate	Less	Blockage in feed valve	Insufficient feed into column may lead to unstable operating conditions	Continuous feed flow monitoring
Compressor	Temperature	More	Inadequate cooling system Recycle stream at an excessively high temperature	Damage to compressor internal mechanism	Temperature control system with alarms and emergency shutdown
	Temperature	Less	Recycle stream at an excessively low temperature	Can lead to condensation of gases at extremely low temperatures and cause corrosion to compressor	Temperature control system with alarms

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	Pressure	More	Malfunctioning control valve Operating above setpoint	Over pressurized vessel can lead to a rupture or explosion	Pressure relief valve High pressure alarms Automatic shut down if pressure exceeds a certain value
	Pressure	Less	Compressor wear Incorrect setpoint	Insufficient compression Reduced recycle flow	Pressure monitoring Routine maintenance to ensure compressor has none to minimal wear
Reboiler	Temperature	More	Steam valve fully open Blockage at reboiler outlet	Overheating of liquid May lead to pressure build up in the column	Temperature control system with alarms Fail-close steam supply valve
	Temperature	Less	Low steam pressure used Heat exchanger fouling	Incomplete vapor generation Poor separation	Temperature control system with alarms Regular cleaning of reboiler tubes
	Pressure	More	Blocked vent line	Over-pressurization of reboiler increases the risk of equipment rupture	Pressure relief valve Pressure monitoring system with alarms
	Pressure	Less	Leak in reboiler shell	Inefficient heat exchanger	Pressure monitoring system with alarms
Condenser	Temperature	More	Cooling water valve closed	Incomplete condensation of vapor	Temperature control system Fail-open valve for cooling water
	Temperature	Less	Cooling water valve fully open	Product may cool past condensation state	Temperature control system with alarms

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	Pressure	More	Vapor overload Condenser tube blockage	Over-pressurized overhead system	Pressure relief system with alarms
	Pressure	Less	Low vapor load Leak in condenser	May affect condensation rate	Pressure monitoring system with alarms
Condensate Steam Heat Exchanger	Temperature	More	Steam flow valve fully open Fouling on cold side of tube and shell exchanger Faulty temperature control system	Over-pressurization may occur and lead to equipment rupture	Temperature control system with alarms
	Temperature	Less	Steam pressure drop Valve closed	Inadequate heat exchange Poor energy recovery	Temperature control system with alarms
	Pressure	More	Blocked pipes Valve malfunction	Equipment rupture Hazard from possible steam leak	Pressure relief valve Regular inspection
	Pressure	Less	Insufficient steam supply Lower pressure used	Poor heat transfer	Pressure monitoring system
	Steam Flow Rate	More	Steam valve fully open	Overheating of process stream	Fail-close valve Flow rate monitor
	Steam Flow Rate	Less	Valve closed Supply line leak	Insufficient heating	Steam flow monitors with alarms
Heat Exchangers	Temperature	More	Excess of hot stream	Inadequate heat transfer	Temperature system with alarms
	Temperature	Less	Excess of cold stream	Incomplete heating	Temperature monitoring with alarms

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	Pressure	More	Blocked pipes Valve malfunction	Equipment rupture Hazard from possible leak	Pressure relief valve Regular inspection
	Pressure	Less	Insufficient supply	Poor heat transfer	Pressure monitoring system
	Flow Rate	More	Valve fully open	Inadequate heat transfer	Flow monitoring with alarms Fail-close valve
	Flow Rate	Less	Valve fully closed Pipe blockage	Overheating of process stream	Flow sensors with alarms

Table E.2: Summary of safety concerns for chemicals of interest and how to safely store and handle each chemical. Safety data regarding reactant chemicals,product, co-product, and by-product is available here.

Species	Flash Point [°C]	Auto-Ignition Temperature [°C]	Explosive Limits [vol%]	Safety Concerns	Storage and Handling
Methanol $[CH_3OH]$[29]	9	440	6-50	Limit of exposure of 200 ppm over an eight-hour shift. Flammable in the presence of open flames, sparks, and smoke.	Store in a well ventilated fireproof area separate from incompatible materials. PPE for handling methanol includes gloves, eye protection, and face protection.
Carbon Monoxide $[CO]$[30]	−191	605	12.5-74.2	Limit of exposure of 25 ppm over and eight-hour shift. Flammable in the presence of open flames, sparks, and smoke. Mixes well with air and creates and explosive mixture. The gas penetrates easily through walls and ceilings. Every year and average of 400 Americans die from carbon monoxide poisoning.	Store in a well ventilated and fireproof area. PPE for handling carbon monoxide includes breathing protection.
Oxygen $[O_2]$[31]	Not Applicable	Not Applicable	Not Applicable	Short term exposure of oxygen at high concentrations can cause effects on the central nervous system,	Store in a cool and fireproof area away from combustibles and reducing agents. PPE

Species	Flash Point [°C]	Auto-Ignition Temperature [°C]	Explosive Limits [vol%]	Safety Concerns	Storage and Handling
				lungs, and eyes. Although oxygen itself is not combustible it enhances combustion of other substances. Keep away from open flames, spark, smoke, and flammables.	for handling oxygen includes safety goggles.
Dimethyl Carbonate $[C_3H_6O_3]$[32]	18	458	4.2-12.9	Creates an explosive mixture with air. Flammable in the presence of open flames, sparks, and smoke. The vapor is heavier than air and can travel along the ground. Short term exposure can lead to mildly irritating to the eyes.	Store in a fireproof well closed area away from oxidants. Store in an area without drain or sewer access. PPE for dimethyl carbonate includes protective gloves and safety goggles.
Carbon Dioxide $[CO_2]$[33]	Not Applicable	Not Applicable	Not Applicable	Limit of exposure of 5000 ppm over an eight-hour shift. Long-term or repeated exposure may have effects on the metabolism. Contact with carbon dioxide in any of this forms can lead to frostbite. Gas is heavier than air and may accumulate an	Store in a fireproof area away with ventilation along the floor. PPE for handling carbon dioxide includes cold-insulating gloves and safety glasses.

Species	Flash Point [°C]	Auto-Ignition Temperature [°C]	Explosive Limits [vol%]	Safety Concerns	Storage and Handling
				cause a deficiency of oxygen.	
Cuprous Chloride [CuCl][20]	Not Applicable	Not Applicable	Not Applicable	Limit of exposure of 1 $\frac{mg}{m^3}$ over an eight-hour shift. Can cause serious eye damage and skin irratation.	Store in a dry, cool, and well ventilated area away from metals and strong oxidizing agents. Keep away from drains and ground water. PPE for cuprous chloride includes face protection, eye protection, gloves, and protective clothes.
Water [H_2O][34]	Not Applicable	Not Applicable	Not Applicable	Not a hazardous substance or mixture.	PPE for water includes safety glasses.
Hydrochloric Acid [HCl][21]	Not Applicable	Not Applicable	Not Applicable	Limit of exposure of 2 ppm over an eight-hour shift. Hydrochloric Acid causes burns by all exposure routes and it is a corrosive material.	Store in a corrosive resistent polypropylene container in a dry, well ventilated area. Keep away from metals, strong oxidizing agents, bases, sodium hypochlorite, amines, fluorine, cyanides, and alkaline. PPE for hydrochloric acid includes face protection, eye protection, gloves, and protective clothes.
Methyl-Isobutyl-Ketone [$C_6H_{12}O$][35]	14	Not Applicable	1.2-8	Limit of exposure of 75 ppm over an eight-hour	Keep away from het, sparks, open flames,

Species	Flash Point [°C]	Auto-Ignition Temperature [°C]	Explosive Limits [vol%]	Safety Concerns	Storage and Handling
				shift. Suspected to cause cancer if inhaled. Confirmed animal carcinogen.	and hot surfaces. Store in a dry and well ventilated area. PPE includes protective clothing and face and eye protection.

Table F.14 : Material properties of chemicals under normal conditions of 25°C and 1 atm.

Species	**Molecular Weight** $[g/mol]$	**Boiling Point** $[°C]$	**Melting Point** $[°C]$	**Density** $[kg/m^3]$
Methanol $[CH_3OH]$[29]	32.0	65	−98	790
Carbon Monoxide $[CO]$[30]	28.0	−191	−205	1.18
Oxygen $[O_2]$[31]	32.0	−183	−218.4	1.35
Dimethyl Carbonate $[C_3H_6O_3]$[32]	90.1	90	3	1070
Carbon Dioxide $[CO_2]$[36]	44.0	−56.6	−78.5	1.84
Cuprous Chloride $[ClCu]$[20]	99.0	1,490	423	4140
Water $[H_2O]$[34]	18.0	100	0	1000
Hydrochloric Acid $[HCl]$[21]	36.5	51	−25	1300
Methyl-Isobutyl-Ketone $[C_6H_{12}O]$[35]	100.2	117-118	−80	801

Table F.15 : Species compatibility with possible plant material

Species	Carbon Steel[37]	Stainless[38]	PTFE-Lined Carbon Steel[39]
Methanol $[CH_3OH]$	A-Excellent	A-Excellent	A-Excellent
Carbon Monoxide $[CO]$	A-Excellent	A-Excellent	A-Excellent
Oxygen $[O_2]$	N/A	N/A	N/A
Dimethyl Carbonate $[C_3H_6O_3]$	N/A	N/A	N/A
Carbon Dioxide $[CO_2]$	A-Excellent	A-Excellent	A-Excellent
Cuprous Chloride $[ClCu]$	N/A	D-Severe Effect	A-Excellent
Water $[H_{20}]$	D-Severe Effect	A-Excellent	A-Excellent
Hydrochloric Acid $[HCl]$	D-Severe Effect	D-Severe Effect	A-Excellent
Methyl-Isobutyl-Ketone $[C_6H_{12}O]$	A-Excellent	B-Good	A-Excellent

Appendix G: MATLAB CODE

G.1: Material Balances

Code that calculated the IO Level and RR Level mole balances. Mass balance checks were also implemented to verify that the resulting balances were correct.

```
function [mole_streams, extents] = solve_mole_streams_psa(spec, constants)

    molar_ratio_CO = spec.molar_ratio_CO;
    molar_ratio_MeOH = spec.molar_ratio_MeOH;
    selectivity = spec.selectivity;
    conversion = spec.conversion;
    dmc_production = constants.molar_production_DMC;

    species = constants.componentdata.specie_indices;
    mm = constants.componentdata.molarmass;

    % IO Level Balances
    mole_streams.FeedMethanol = MaterialStream(mm);
    mole_streams.FeedOxygen = MaterialStream(mm);
    mole_streams.FeedCarbonMonoxide = MaterialStream(mm);
    mole_streams.ProductDMC = MaterialStream(mm);
    mole_streams.ProductWater = MaterialStream(mm);
    mole_streams.ProductCarbonDioxide = MaterialStream(mm);

    % RR Level Balances
    mole_streams.RecycleMethanol = MaterialStream(mm);
    mole_streams.RecycleGases = MaterialStream(mm);
    mole_streams.ReactorFeedLiquid = MaterialStream(mm);
    mole_streams.ReactorFeedGas = MaterialStream(mm);
    mole_streams.ReactorFeed = MaterialStream(mm);
    mole_streams.PrePSARecycleGas = MaterialStream(mm);

   % Calculate the IO and RR Balances, which were solved simulatanously using Mathimatica
    % FMeOH
    FMeOH = 2 .* dmc_production;

    % FCO
    FCO = dmc_production ./ (selectivity);

    % FO2
    FO2 = dmc_production ./ (2.*selectivity);

    % RMeOH
        RMeOH  =  ((4.*selectivity.*conversion  +  molar_ratio_MeOH).*dmc_production)./
(2.*selectivity.*conversion);

    % RCO
  RCO = ((molar_ratio_CO - 2.*conversion).*dmc_production)./(2.*selectivity.*conversion);

    % RO2
    RO2 = ((1 - conversion).*dmc_production)./(2.*selectivity.*conversion);
```

```
    % PCO2
    PCO2 = dmc_production.*(1 - selectivity)./selectivity;

    % PH2O
    PH2O = dmc_production;

    % IO2
    IO2 = dmc_production./(2 .* selectivity .* conversion);

    % IMeOH
    IMeOH = (molar_ratio_MeOH.*dmc_production)./(2.*selectivity.*conversion);

    % ICO
    ICO = (molar_ratio_CO.*dmc_production)./(2.*selectivity.*conversion);

    % Provide the extents of reaction
    extents = [dmc_production, PCO2];

    % Assign the streams
    ProductDMC_comp = zeros(1,6);
    ProductDMC_comp(species.dmc) = 1;
         mole_streams.ProductDMC   =  mole_streams.ProductDMC.setByMolar(dmc_production,
ProductDMC_comp);

    FMeOH_comp = zeros(1,6);
    FMeOH_comp(species.methanol) = 1;
    mole_streams.FeedMethanol = mole_streams.FeedMethanol.setByMolar(FMeOH, FMeOH_comp);

    FO2_comp = zeros(1,6);
    FO2_comp(species.oxygen) = 1;
    mole_streams.FeedOxygen = mole_streams.FeedOxygen.setByMolar(FO2, FO2_comp);

    FCO_comp = zeros(1,6);
    FCO_comp(species.carbon_monoxide) = 1;
      mole_streams.FeedCarbonMonoxide  =  mole_streams.FeedCarbonMonoxide.setByMolar(FCO,
FCO_comp);

    PH2O_comp = zeros(1,6);
    PH2O_comp(species.water) = 1;
    mole_streams.ProductWater = mole_streams.ProductWater.setByMolar(PH2O, PH2O_comp);

    Pgas_comp = zeros(1,6);
    Pgas_comp(species.carbon_dioxide) = 1;
   mole_streams.ProductCarbonDioxide = mole_streams.ProductCarbonDioxide.setByMolar(PCO2,
Pgas_comp);

    RMeOH_comp = zeros(1,6);
    RMeOH_comp(species.methanol) = 1;
         mole_streams.RecycleMethanol   =  mole_streams.RecycleMethanol.setByMolar(RMeOH,
RMeOH_comp);
```

```
    Rgas = RO2 + RCO;
    RGas_comp = zeros(1,6);
    RGas_comp(species.oxygen) = RO2./Rgas;
    RGas_comp(species.carbon_monoxide) = RCO./Rgas;
    mole_streams.RecycleGases = mole_streams.RecycleGases.setByMolar(Rgas, RGas_comp);

    ReactorFeedGas = IO2 + ICO;
    ReactorFeedGas_comp = zeros(1,6);
    ReactorFeedGas_comp(species.oxygen) = IO2./ReactorFeedGas;
    ReactorFeedGas_comp(species.carbon_monoxide) = ICO./ReactorFeedGas;
    mole_streams.ReactorFeedGas = mole_streams.ReactorFeedGas.setByMolar(ReactorFeedGas,
ReactorFeedGas_comp);

    ReactorFeed = IO2 + ICO + IMeOH;
    ReactorFeed_comp = zeros(1,6);
    ReactorFeed_comp(species.oxygen) = IO2./ReactorFeed;
    ReactorFeed_comp(species.carbon_monoxide) = ICO./ReactorFeed;
    ReactorFeed_comp(species.methanol) = IMeOH./ReactorFeed;
          mole_streams.ReactorFeed  =  mole_streams.ReactorFeed.setByMolar(ReactorFeed,
ReactorFeed_comp);

    PrePSARecycleGas = PCO2 + RO2 + RCO;
    PrePSARecycleGas_comp = zeros(1,6);
    PrePSARecycleGas_comp(species.oxygen) = RO2./PrePSARecycleGas;
    PrePSARecycleGas_comp(species.carbon_monoxide) = RCO./PrePSARecycleGas;
    PrePSARecycleGas_comp(species.carbon_dioxide) = PCO2./PrePSARecycleGas;
                                                mole_streams.PrePSARecycleGas          =
mole_streams.PrePSARecycleGas.setByMolar(PrePSARecycleGas, PrePSARecycleGas_comp);

    ReactorFeedLiquid_comp = zeros(1,6);
    ReactorFeedLiquid_comp(species.methanol) = 1;
      mole_streams.ReactorFeedLiquid  =  mole_streams.ReactorFeedLiquid.setByMolar(IMeOH,
ReactorFeedLiquid_comp);
end
```

G.2: Reactor Design and Kinetics

Solves the reactor design equation using the closed mole balance formulas at the reactor feed. This is returned as a function of theoretical volume. The reactor's conversion and selectivity is then computed.

The selectivity is iterated to ensure the mole balance is valid. The selectivity used to calculated at the reactor feed must match the calculated reactor selectivity.

General PFR reactor ODE. As we have a adiabatic reactor, we must account for the reactor energy balance for accurate prediction of the reactor volume as a function of conversion. Heat capacity estimates are handled using the Allen-Lee-Dipper approximation as a function of temperature. Individual parameters are included inline with the heat capacity data. For the temperature range were are considering, the heat of reactions were found to be weakly a function of temperature and hence are considered constant.

```
function [f,J] = solve_two_phase_CSTR_balance(f0, N0, V, T, K, Q, constants)
    nu = constants.componentdata.stoich_matrix;
    species = constants.componentdata.specie_indices;
```

```
N = f0(1:end-1);
phi = f0(end);

Ntot = sum(N);
Ntot0 = sum(N0);
z = N ./ Ntot;
z0 = N0 ./ Ntot0;

Ltot = (1 - phi)*Ntot;
x = z ./ (1 + phi.*(K - 1));
L = x .* Ltot;

Ltot0 = (1 - phi)*Ntot0;
x0 = z0 ./ (1 + phi.*(K - 1));
L0 = x0 .* Ltot0;

%Q = calculate_volumetric_flow(T, Ltot, x, constants);

% Rate constants
R = 8.314;
k1 = 1.6e11 .* exp(-104600 ./ (R .* T));  % returns k1(T)
k1_tilde = k1 / (Q^2.5);

k2 = 6.7e12 .* exp(-98750 ./ (R .* T));
k2_tilde = k2 / (Q^0.5);

r1 = k1_tilde * N(species.methanol)^2 * N(species.oxygen)^0.5;
r2 = k2_tilde * N(species.oxygen)^0.5;

% mol/(L*s)
r = [r1, r2];

% Residuals
% L * mol/(L*s) -> mol / s
R = V * (r * nu);  % 1x6
f = zeros(7, 1);
% mol / s
f(1:6) = N0 - N + R;

% Rachford-Rice residual
f(7) = sum((K - 1) .* z ./ (1 + phi * (K - 1)));

% Jacobian
J = zeros(7, 7);

t1 = V*k1_tilde*N(species.methanol)*N(species.oxygen)^0.5;
J(1,1) = -1 - 4*t1;
J(2,1) = -t1;
J(3,1) = -2*t1;
J(4,1) = 2*t1;
J(5,1) = 2*t1;
```

```
    t2 = V*k1_tilde*N(species.methanol)^2;
    t3 = N(species.oxygen)^0.5;
    J(1,2) = -t2/t3;
    J(2,2) = -1 - (V*k2_tilde + t2)/(4*t3);
    J(3,2) = -(V*k2_tilde + t2)/(2*t3);
    J(4,2) = t2/(2*t3);
    J(5,2) = t2/(2*t3);
    J(6,2) = k2_tilde*V/(2*t3);

    J(3,3) = -1;
    J(4,4) = -1;
    J(5,5) = -1;
    J(6,6) = -1;

    for j = 1:6
        comp = false(1,6);
        comp(j) = true;
        remaining_comp = ~comp;

                J(7,j) = -(sum(((K(remaining_comp) - 1).*N(remaining_comp))./(1 +
phi*(K(remaining_comp) - 1))) ...
                    - ((K(comp) - 1).*sum(N(remaining_comp)))/(1+phi*(K(comp)-1)));
    end
    J(7,1:6) = J(7,1:6) ./ (Ntot.^2);

    J(7,7) = -sum((((K-1).^2).*N)./(1+((K-1).*phi).^2))/Ntot;

    % f = real(f);
    % J = real(J);
end
```

G.3: Optimization using NPV

In order to obtain the best NPV, we using a objective function that returns the NPV given select optimization parameters, which includes the inlet reactor feed temperature T_i, molar ratio MR, and purge fraction of hydrogen y_{PH}.

$$\text{NPV} = f(x, y_{\text{PH}}, T_i) \tag{G.1}$$

This is done using the `fmincon` function. This is a minimization function, but we can seek for a maximization by considering the negative of the objective and then taking the negative of its minimization value. The objective function in this case is called `objective_npv`. Other minor variables were briefly simulated, include the distillation column pressures, but were generally excluded from the typical optimization setup.

MATLAB's `fmincon` function also allows upper and lower bounds, which were used to limit that the

$$\begin{aligned} g(x, y_{\text{PH}}, T_i) &= -f(x, y_{\text{PH}}, T_i) \\ [-x, -y_{\text{PH}}, -T_i] &= \overset{\text{fmincon}}{\max}(-f(x, y_{\text{PH}}, T_i)) \end{aligned} \tag{G.2}$$

```
function sol = optimize_gas_sep(optimizer, use_parallel, constants)
    function NPV = objective_NPV(x)
        input_sol = generate_sol_psa(x);
        result = generate_reactor_case(input_sol, "gas_CO2_sep", constants);
```

```
        if ~isnan(result.NPV)
            NPV = -result.NPV;
        else
            NPV = realmax("double")*1e-3;
        end
    end

    %     tau     MR_CO      MR_MeOH
    x0 = [5*60    25,        10     ];
    lb = [1*60    24.0001    1      ];
    ub = [Inf     100        100    ];

    sol = optimizer(x0, lb, ub, @objective_NPV, use_parallel);
end
```

G.4: Plant Economic and Plant

```
function econ_out = plant_economics(specs, mole_streams, plant_parameters, sub_sep_fun,
constants)

    % Equipment Costing
    [isbl, opex] = equipment_costing(mole_streams, plant_parameters, specs, sub_sep_fun,
constants);

    ISBL = sum(table2array(struct2table(isbl)));
    OPEX = sum(table2array(struct2table(opex)));

    % Cost/Revenue from Reactants/Products
    [rcosts, pcosts] = flow_costing(mole_streams, constants);

    reactantCosts = zeros(1, constants.econ.plant_life);
    productRevenue = zeros(1, constants.econ.plant_life);

    for i = 1:constants.econ.plant_life
        if i >= constants.econ.price_feed_increase_start_time && ...
           i <= constants.econ.price_feed_increase_start_time + ...
                constants.econ.price_feed_increase_time
            reactantCosts(i) = rcosts * price_feed_increase;
        else
            reactantCosts(i) = rcosts;
        end

        if i >= constants.econ.price_product_decrease_start_time && ...
           i <= constants.econ.price_product_decrease_start_time + ...
                constants.econ.price_product_decrease_time
            productRevenue(i) = pcosts * price_product_decrease;
        else
            productRevenue(i) = pcosts;
        end
    end

    AGS_cost = constants.econ.AGS_rate .* productRevenue;
```

```
% Fixed capital investment
FCI = constants.econ.fci_multipler .* ISBL;

% Working capital = 2 months of feedstock cost (first year)
WC = (2/12) .* reactantCosts(1);  % Note: WC is based on year 1 cost
SU = 0.1 .* FCI;                  % Startup cost = 10% of FCI

% Per-year total manufacturing cost
COM = OPEX + reactantCosts + AGS_cost;  % all 1-by-N vectors

% Salvage value
salvage = FCI * constants.econ.salvage_rate;
costbasis = FCI + SU - salvage;

% Depreciation schedule
construction_years = round(constants.econ.construction);
depreciation = zeros(constants.econ.plant_life, 1);
depreciation(construction_years+1 : construction_years+constants.econ.dep_sched) = ...
    costbasis / constants.econ.dep_sched;

% Cash Flow array
CF = zeros(constants.econ.plant_life, 1);

% Construction phase
for i = 2 : construction_years + 1
    CF(i) = -FCI / construction_years;
end

% Last year of construction: include SU and WC
CF(construction_years + 1) = CF(construction_years + 1) - WC - SU;

% Operating phase
for i = (construction_years + 2) : constants.econ.plant_life - 1
    rev = productRevenue(i);
    cost = COM(i);
    dep = depreciation(i);
    CF(i) = (rev - cost - dep) * (1 - constants.econ.tax_rate) + dep;
end

% Final year: include salvage and return of WC
i = constants.econ.plant_life;
rev = productRevenue(i);
cost = COM(i);
dep = depreciation(i);
CF(i) = (rev - cost - dep + salvage) * (1 - constants.econ.tax_rate) + dep + WC;

% Discounted Cash Flow
DCF = zeros(constants.econ.plant_life, 1);
for k = 1:constants.econ.plant_life
    DCF(k) = CF(k) / ((1 + constants.econ.enterprise_rate)^(k - 1));
end
```

```
    % Generate the cumulative discounted cash flows, NPV, and EP
    econ_out.isbl = isbl;
    econ_out.isbl_total = ISBL;
    econ_out.opex = opex;
    econ_out.opex_total = OPEX;
    econ_out.fci = FCI;
    econ_out.tci = FCI + WC + SU;
    econ_out.wc = WC;
    econ_out.su = SU;
    econ_out.reactants_costs = reactantCosts;
    econ_out.product_revenue = productRevenue;
    econ_out.opex_ags = AGS_cost;
    econ_out.cdcf = cumsum(DCF)';
    econ_out.npv = econ_out.cdcf(end) + DCF(end);
    econ_out.ep = productRevenue - reactantCosts;
end
```

G.5: Tornado Plot

```
function tornado_plot(baseValue, paramNames, lowValues, highValues)
%TORNADO_PLOT Create a tornado plot for sensitivity analysis with labels
%
% Inputs:
%   baseValue   - scalar baseline value (e.g., baseline NPV)
%   paramNames  - cell array of parameter names
%   lowValues   - vector of values when each parameter is perturbed low
%   highValues  - vector of values when each parameter is perturbed high

    positive_color = [0.3 0.8 0.3];
    negative_color = [1 0.3 0.3];

    % Ensure input validity
    assert(length(paramNames) == length(lowValues) && ...
           length(paramNames) == length(highValues), ...
           'All input vectors must be the same length');

    % Compute impacts relative to base
    lowImpact = lowValues - baseValue;
    highImpact = highValues - baseValue;
    totalImpact = max(abs([lowImpact; highImpact]), [], 1);

    % Sort by absolute impact
    [~, idx] = sort(totalImpact, 'descend');
    paramNames = paramNames(idx);
    lowImpact = lowImpact(idx);
    highImpact = highImpact(idx);
    lowValues = lowValues(idx);
    highValues = highValues(idx);

    % Prepare y-axis indices
    n = length(paramNames);
    y = 1:n;
```

```
    % Track limits for autoscaling
    x_min = 0;
    x_max = 0;

    % Plot setup
    fig = figure;
    fig.Units = 'inches';
    fig.Position(3:4)=[5,3];
    textsize = 10;

    hold on;

    for i = 1:n
        % Determine colors
      lowImpactColor = positive_color * (lowImpact(i) >= 0) + negative_color * (lowImpact(i)
< 0);
           highImpactColor = positive_color * (highImpact(i) >= 0) + negative_color *
(highImpact(i) < 0);

        % Bar plots
        barh(y(i), lowImpact(i), 0.4, 'FaceColor', lowImpactColor, 'EdgeColor', 'none');
       barh(y(i), highImpact(i), 0.4, 'FaceColor', highImpactColor, 'EdgeColor', 'none');

        % Text labels outside of bars
        if lowImpact(i) < 0
            xpos = lowImpact(i) - 0.01 * abs(lowImpact(i));
            halign = 'right';
        else
            xpos = lowImpact(i) + 0.01 * abs(lowImpact(i));
            halign = 'left';
        end
        text(xpos, y(i), sprintf('%.1f', lowValues(i)), ...
              'HorizontalAlignment', halign, 'VerticalAlignment', 'middle', 'FontSize',
textsize);

        if highImpact(i) < 0
            xpos = highImpact(i) - 0.01 * abs(highImpact(i));
            halign = 'right';
        else
            xpos = highImpact(i) + 0.01 * abs(highImpact(i));
            halign = 'left';
        end
        text(xpos, y(i), sprintf('%.1f', highValues(i)), ...
              'HorizontalAlignment', halign, 'VerticalAlignment', 'middle', 'FontSize',
textsize);

        % Track min/max for x-axis scaling
        x_min = min([x_min, lowImpact(i), highImpact(i), xpos]);
        x_max = max([x_max, lowImpact(i), highImpact(i), xpos]);
    end

    % Add baseline line and horizontal label
```

```
    xline(0, '--k');
    text(0, n + 0.6, sprintf('Base = %.1f', baseValue), ...
         'HorizontalAlignment', 'center', 'VerticalAlignment', 'bottom', ...
         'FontSize', textsize);

    % Axes formatting
   set(gca, 'YTick', y, 'YTickLabel', paramNames, 'YDir', 'reverse', 'FontSize', textsize);
    xlabel(sprintf('Change from baseline (%.1f)', baseValue), 'FontSize', textsize);
    title('Tornado Plot', 'FontSize', textsize, 'FontWeight', 'bold');
    grid on;
    box on;

    % Set tight but padded x limits
    margin = 0.08 * (x_max - x_min);
    xlim([x_min - margin, x_max + margin]);

    hold off;
end
```

G.6: Monte Carlo Simulation

```
% Get base NPV

base_npv = econ_optimized.npv / 1e6;  % in millions

% Simulation parameters

%N = 10000;  % Large number for HPC
npv_results = NaN(N, 1);  % Preallocate

% Define bounds for triangular distributions
construction_min = 2; construction_mode = 3; construction_max = 6;
toluene_min = 270; toluene_mode = 300; toluene_max = 390;
construction_min = 2; construction_mode = 3.0; construction_max = 5.0;
%conversion_min = 0.6; conversion_mode = optimized_spec.x; conversion_max = 0.9;

% Open a parallel pool (let's match # cpus you request)
% parpool('local', str2double(getenv('SLURM_CPUS_PER_TASK')));

parfor k = 1:N
    try
        constants_sim = get_constants();

        % Sample uncertain parameters
        benzene_price = triangular_sample(benzene_min, benzene_mode, benzene_max, 1);
        toluene_price = triangular_sample(toluene_min, toluene_mode, toluene_max, 1);
             construction_time  =  triangular_sample(construction_min,  construction_mode,
construction_max, 1);
      %conversion = triangular_sample(conversion_min, conversion_mode, conversion_max, 1);

        % Update constants with sampled values
        constants_sim.econ.benzeneprice = benzene_price;
        constants_sim.econ.toluenecost = toluene_price;
```

```
        constants_sim.econ.construction = construction_time;
        optimized_spec_local = optimized_spec;
        %optimized_spec_local.x = conversion;

        % Solve plant model
        econ_sim = solve_HDA_plant(optimized_spec_local, constants_sim);
        npv_results(k) = econ_sim.npv / 1e6;
    catch
        npv_results(k) = NaN;
    end
end

delete(gcp('nocreate'));

% Clean and save results
npv_results = npv_results(~isnan(npv_results) & npv_results > 0);
writematrix(npv_results, "monte_results.csv");

fig1 = figure;
hold on;
xline(base_npv, 'LineWidth', 2, 'Color', 'black','LineStyle', '--')
histogram(npv_results, 80, 'FaceColor', '#66c2a5');
xlabel('NPV [$MM]');
ylabel('Frequency');
hold off;
savefig(fig1, "histogram.fig");

% --- CDF Plot with Shaded 95% Prediction Interval and Annotations ---
fig2 = figure;
sorted_npv = sort(npv_results);
cdf = linspace(0, 1, length(sorted_npv));

% Interpolate to smooth
npv_interp = linspace(min(sorted_npv), max(sorted_npv), 500); % 500 smooth points
cdf_interp = interp1(sorted_npv, cdf, npv_interp, 'pchip');

% First, plot the shaded 95% interval
P2_5 = prctile(npv_results, 2.5);
P97_5 = prctile(npv_results, 97.5);

% Create shaded area between 2.5% and 97.5% bounds
x_shade = [P2_5, P97_5, P97_5, P2_5];
y_shade = [0, 0, 1, 1];
fill(x_shade, y_shade, [0.9 0.9 0.9], 'EdgeColor', 'none', 'FaceAlpha', 0.5); % Light
gray shade

hold on;

% Plot the empirical CDF
plot(npv_interp, cdf_interp, 'b-', 'LineWidth', 2, 'DisplayName', 'Empirical CDF');

% Find cumulative probability of the base case NPV
```

```
[~, idx_base] = min(abs(sorted_npv - base_npv));
base_cdf_value = cdf(idx_base);

% Plot base NPV case
scatter(base_npv, base_cdf_value, 'red', 'filled')

hold off;
xlabel('NPV [$MM]');
ylabel('Cumulative Probability');
legend('Location', 'southeast');
grid on;

savefig(fig2, "CDF.fig");

writematrix(x_shade, "quantiles.csv");
```

Appendix H: Team Member Work Statements

Team Member Work Statement

Name of team member here Lauren Bajo-Smith

State here what you contributed to the design project and report

ASPEN HYSYS – w/ Ashley
Economic Design Decisions + Calculation
Sensitivity Analysis, majority of Code
written report
+
appendices

Name of team member here
Ashley Uriostegui

State here what you contributed to the design project and report

ASPEN HYSYS – w/ Lauren
HAZOP analysis
Safety summary and tables in appendix
Safety hazards part of report and pie charts

Name of team member here
Forrest Ramirez

State here what you contributed to the design project and report

MATLAB conceptual design, including reactor modeling, Level 1-3 molar flowrates, implementation of economic/npv calculations, and optiminization algorithm. Key figure plots. Typst/LaTex formatting.

Print Name and Sign: Lauren Bajo-Smith Lauren Bajo-Smith Date: 3-12-25

Print Name and Sign Ashley Uriostegui Ashley Uriostegui Date: 3-12-25

Print Name and Sign Forrest Ramirez FRA Date: 3-12-25

Rating of Team Members for Design Project

Please rate each group member's contributions in the categories below:
1-2 - unsatisfactory, 3 - acceptable/adequate, 4 – very good, 5 - excellent
Each member fills out one form and signs the bottom.

Name :	1) Ashley	2) Lauren	3) Forrest
Quality of work presented	5	5	5
Quantity of work performed	5	5	5
Effort	5	5	5
Punctuality (meetings and deadlines)	5	5	5
Knowledge of design methods	4	4	4
Class attendance	5	5	5
Communication	5	5	5

Do you feel that each member of the group deserves the same grade? If not, who does not and why?

Yes! Everyone worked very hard. Amazing Team!

It's important to note that differences in performance will not necessarily affect individual grades; however, large discrepancies may result in differences in grades.

Additional comments:

Print Name and Sign: Forrest Ramirez FAR Date: 3-12-25

Rating of Team Members for Design Project

Please rate each group member's contributions in the categories below:
1-2 - unsatisfactory, 3 - acceptable/adequate, 4 – very good, 5 - excellent
Each member fills out one form and signs the bottom.

Name : Lauren	1) Lauren	2) Forrest	3) Ashley
Quality of work presented	5	5	5
Quantity of work performed	5	5	5
Effort	5	5	5
Punctuality (meetings and deadlines)	5	5	5
Knowledge of design methods	4	4	4
Class attendance	5	5	5
Communication	5	5	5

Do you feel that each member of the group deserves the same grade? If not, who does not and why?

Yes, solid team! I think we work well together.

It's important to note that differences in performance will not necessarily affect individual grades; however, large discrepancies may result in differences in grades.

Additional comments:

Print Name and Sign Lauren Bajo Smith Date: 3-12-25

Rating of Team Members for Design Project

Please rate each group member's contributions in the categories below:
1-2 - unsatisfactory, 3 - acceptable/adequate, 4 – very good, 5 - excellent
Each member fills out one form and signs the bottom.

Name : Ashley	1) Ashley	2) Lauren	3) Forrest
Quality of work presented	5	5	5
Quantity of work performed	5	5	5
Effort	5	5	5
Punctuality (meetings and deadlines)	5	5	5
Knowledge of design methods	4	4	4
Class attendance	5	5	5
Communication	5	5	5

Do you feel that each member of the group deserves the same grade? If not, who does not and why?

Yes, everyone deserves the same grade
Everyone contributed in their own way

Great team!

It's important to note that differences in performance will not necessarily affect individual grades; however, large discrepancies may result in differences in grades.

Additional comments:

Print Name and Sign: Ashley Uriostegui Ashley Uriostegui Date: 3-12-25